

A Comprehensive Survey of Website Fingerprinting Attacks and Defenses in Tor: Advances and Open Challenges

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The Tor network provides users with strong anonymity by routing their internet traffic through multiple relays. While Tor encrypts traffic and hides IP addresses, it remains vulnerable to traffic analysis attacks such as the website fingerprinting (WF) attack, achieving increasingly high fingerprinting accuracy even under open-world conditions. In response, researchers have proposed a variety of defenses, ranging from adaptive padding, traffic regularization, and traffic morphing to adversarial perturbation, that seek to obfuscate or reshape traffic traces. However, these defenses often entail trade-offs between privacy, usability, and system performance. Despite extensive research, a comprehensive survey unifying WF datasets, attack methodologies, and defense strategies remains absent. This paper fills that gap by systematically categorizing existing WF research into three key domains: datasets, attack models, and defense mechanisms. We provide an in-depth comparative analysis of techniques, highlight their strengths and limitations under diverse threat models, and discuss emerging challenges such as multi-tab browsing and coarse-grained traffic features. By consolidating prior work and identifying open research directions, this survey serves as a foundation for advancing stronger privacy protection in Tor. We summarize representative WF attack and defense studies, along with their applied datasets, on GitHub (<https://github.com/WF-Attack-and-Defense/Awesome-WF-Security>).

Index Terms—Anonymity, website fingerprinting, WF attack, WF defense, machine learning, deep learning.

I. INTRODUCTION

The Tor network is a decentralized, volunteer-operated system designed to provide website browsing anonymity. Tor applies sophisticated onion routing through a series of relay nodes and encryption mechanisms to anonymize traffic. Its architecture is built to protect users from surveillance, traffic analysis, and censorship by hiding their identity and online activity. Thus, Tor makes adversaries difficult to trace the origin or destination of the data [1]–[4].

However, Tor’s network traffic can still be susceptible to traffic analysis techniques, including website fingerprinting (WF) attacks [5]–[7]. By exploiting patterns in encrypted traffic, these attacks aim to expose the websites a user is visiting. Distinctive patterns in packet size [8]–[10], timing [11]–[13], and flow direction [12], [14] can still leak identifying information about the websites being accessed. In response, a variety of defense mechanisms [15]–[17] have been developed to obscure or randomize traffic patterns, thereby reducing

the risk of identification. These strategies include traffic obfuscation [18], [19], adversarial perturbation [20], [21], and website morphing [22]–[24]. While effective in mitigating fingerprinting attempts, such defenses must carefully balance privacy protection with performance to ensure they do not degrade the user experience.

A. Motivation

Despite the growing significance of WF attack and defense approaches, a systematic and comprehensive survey of the field remains lacking in academia. Existing surveys primarily focus on WF attacks and defenses using machine learning (ML) models [25]–[27]. However, these works provide limited insights into the design of models in critical aspects such as concept drift, single-tab versus multi-tab browsing scenarios, and dynamic adversarial capabilities. Moreover, while previous surveys [27], [28] introduce several defense mechanisms, their taxonomies are neither systematic nor sufficiently granular. For instance, Chao *et al.* [28] categorize WF attacks and defenses within anonymity networks based on different protocol layers of the onion route. However, a gap remains in understanding the comparative effectiveness of various techniques under diverse threat models and defense scenarios. Existing work fails to adequately categorize defenses according to design principles (*e.g.*, obfuscation, padding, traffic morphing) or threat model alignment, limiting their utility for comparative evaluation. Furthermore, datasets in shaping WF research are not well justified. These surveys fail to categorize defenses based on factors such as design principle (*e.g.*, obfuscation, padding, traffic morphing) or threat model alignment adequately. For example, the existing survey [27] discusses dataset features and assumptions, but does not examine the availability, diversity, and limitations of existing datasets in depth. Therefore, a systematic review of WF attacks and defenses in Tor is imperative. Unlike existing surveys, our work provides an in-depth review of available datasets and examines their features to guide dataset selection for future research. Building on this foundation, we present a comprehensive review of WF attack approaches and categorize defense strategies according to their underlying mechanisms, highlighting their respective strengths and weaknesses against state-of-the-art (SOTA) attack models.

B. Taxonomies and Our Contributions

In this work, we categorize Tor-based WF security research into three categories: dataset, WF attacks, and WF defenses.

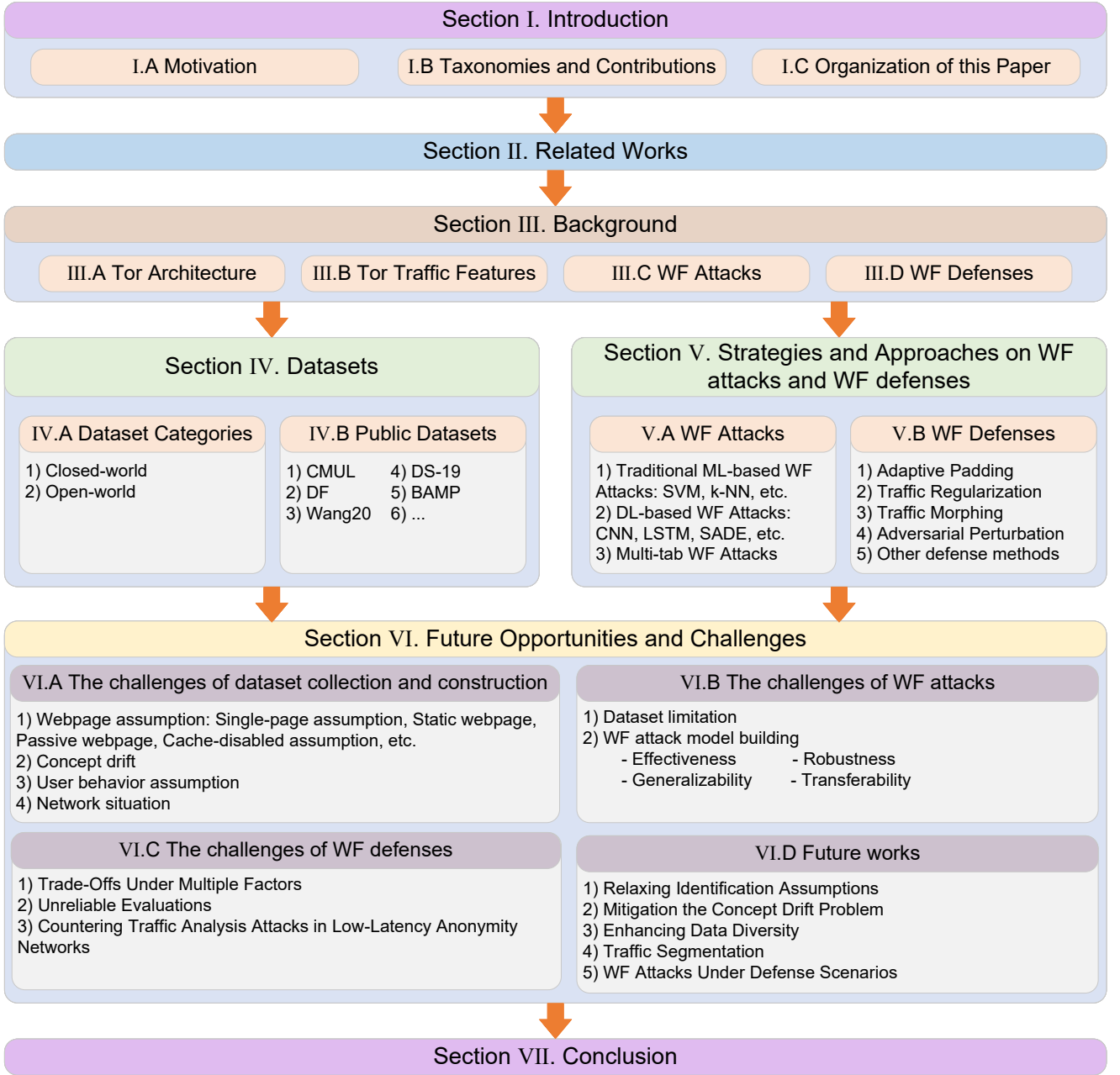


Fig. 1: Structure of the survey. Website fingerprinting security can be categorized into three major categories based on the roles of existing works: dataset, WF attacks, and WF defenses. For each major category, there are threat models associated with it.

1) *Datasets*: Labeled datasets mapping traffic traces to websites are required in the SOTA WF attacks [29]–[33], which usually employ supervised learning techniques. The effectiveness of existing ML models largely depends on the quality, diversity, and scale of the datasets used. Therefore, the development of realistic, diverse, and up-to-date datasets is essential for advancing research in this field and ensuring the reliability of experimental results [34].

There are two types of datasets: closed-world and open-world. A closed-world dataset assumes that the user visits only a predefined set of websites (e.g., the top 100 sites from Alexa), and both the attacker and defender know this set in advance [35]. The closed-world dataset is better suited for

studying the upper bound of attack performance in a controlled setting, as it facilitates comparisons between different WF models under ideal conditions. However, closed-world evaluation overlooks the fact that users can visit websites outside the training set of an ML model. In contrast, an open-world dataset contains a smaller set of monitored websites (the attacker is interested in) and a much larger set of unmonitored websites representing background traffic. In this setting, the attacker must distinguish between monitored and unmonitored sites. Open-world datasets more accurately reflect real-world usage, allowing researchers to evaluate the effectiveness of attacks and defenses under more realistic assumptions [9], [36].

To help researchers select the appropriate dataset for their

experiments and guide them in collecting their own, it is essential to provide a detailed overview of existing datasets, including their characteristics and intended use cases. By understanding the differences between closed-world and open-world datasets, the scope of monitored versus unmonitored sites, and the network conditions under which data was collected, researchers can make informed decisions about which datasets best suit their specific research goals. Moreover, offering guidelines on how to collect traffic traces, such as recommended tools, ethical considerations, labeling practices, and data preprocessing steps, can empower new researchers to build reliable and realistic datasets tailored to their own WF attack or defense studies. This not only enhances the reproducibility and relevance of future work but also helps bridge current gaps in the field by encouraging the development of more diverse and updated datasets.

2) *WF attacks*: The WF attacks section introduces the existing website fingerprinting attacks by providing a comprehensive overview of the various techniques that have been developed over time. This includes a detailed discussion of the underlying models used, ranging from traditional ML algorithms such as k-Nearest Neighbors (k-NN) [9], [33], [35] and Support Vector Machines (SVM) [4], [35], [37] to more advanced DL approaches such as Convolutional Neural Networks (CNN) [13], [38]–[42] and Transformer-based architectures [30], [43], [44]. The section also covers the process of feature selection, highlighting which traffic characteristics, such as packet sizes, directions, timing, and burst patterns, are most informative for classification. Additionally, the performance of different attacks is compared using standard evaluation metrics, such as accuracy, precision, and recall, across both closed-world and open-world scenarios. This comparison can help illustrate the evolution of WF attacks and provide insight into which methods are most effective under various conditions.

3) *WF defenses*: The WF defenses section introduces the existing techniques designed to counter website fingerprinting attacks, with a particular focus on a range of key defense mechanisms. Among these, adaptive padding is a dynamic approach that injects dummy traffic based on live traffic patterns to conceal burst and packet size features, striking a balance between privacy and bandwidth overhead. Traffic morphing attempts to transform the traffic of one website to resemble that of another, thereby confusing fingerprinting classifiers by altering key statistical features [9], [23], [24], [45]. Adversarial perturbation techniques borrow principles from adversarial ML to proactively disrupt attack models [46]–[48], where minor, intentional modifications to network traffic are used to mislead ML models. Each method is discussed in terms of its effectiveness, overhead, implementation complexity, and suitability for real-world Tor systems. This in-depth review will help readers understand the current state of WF defenses and the challenges that remain in developing robust, low-cost, and scalable solutions.

C. Organization of this Survey Paper

The structure of the survey is shown in Fig. 1. Section II reviews prior taxonomies and introduces a more comprehensive

and detailed classification. Section III provides background on Tor architecture, traffic features used in WF, and related security research. Section IV discusses dataset categories and highlights representative public datasets commonly used in current research. Section V presents a comprehensive survey of WF attack and defense studies. In Section VI, we examine the technical bottlenecks facing WF attacks and defenses, to encourage further research in advancing WF security. Finally, Section VII summarizes and concludes the paper.

II. RELATED WORKS

In this section, we summarize the contributions and limitations in existing WF-related surveys as shown in TABLE I. Existing surveys on WF attacks and defenses either focus solely on WF attacks or exclusively on the application of deep learning techniques. To the best of our knowledge, this is the first work to offer a comprehensive and detailed summary and analysis of both WF attacks and defenses, along with their associated datasets.

Basyoni *et al.* [49] explored the adversarial models of traffic analysis attacks on Tor in 2020. The authors examine traffic analysis attacks based on the adopted adversary model and its alignment with Tor’s threat model. However, the paper reviews only ten traffic analysis attacks from the perspective of the adopted adversary model, examining the effectiveness of each within the context of the adopted adversary model and Tor’s threat assumptions. Moreover, this paper lacks an extensive introduction to the WF attack, thus failing to provide a comprehensive landscape of such attacks. Later, Karunanayake *et al.* [50] surveyed de-anonymization attacks, categorizing them based on the components involved, including entry and exit routers, onion proxy, side channels, and hybrid approaches. However, this survey paper classifies WF attacks under side channels, a categorization that lacks standardization and fails to capture the nature of these attacks accurately. This classification may create ambiguity in distinguishing WF attacks from other de-anonymization techniques. Aminuddin *et al.* [27] provided a comprehensive survey of WF attacks on the Tor network in 2023. This paper systematically reviews existing WF techniques and categorizes them based on five primary aspects: threat model, victim target, website realm, traffic feature, and traffic classifier. This work also critically examines nine key assumptions that limit the real-world applicability of WF techniques, such as closed-world settings, static content, and passive webpage assumptions. However, the survey focuses almost exclusively on attack methodologies, leaving website fingerprinting defenses outside its scope. Moreover, its discussion of datasets is relatively limited, although dataset assumptions and realism are crucial factors that affect the performance of WF attacks. A deeper integration of dataset considerations with attack and defense methodologies would have provided a more balanced perspective.

Liu *et al.* [25] presented a survey on WF attacks and defenses, with a primary focus on deep learning (DL) paradigms. Their work reviews the architecture, efficacy, and computational overhead of DL-based models in WF security. However, their study centers predominantly on DL methodologies rather

TABLE I: The summary of related work.

Publication	Year	Main Focus	Include WF attacks	Include WF defenses	Include Dataset
Basyoni <i>et al.</i> [49]	2020	Traffic Analysis Attacks on Tor.	●	○	○
Karunanayake <i>et al.</i> [50]	2021	De-Anonymisation Attacks on Tor.	●	○	○
Aminuddin <i>et al.</i> [27]	2023	Analysis on Tor-based WF techniques and assumptions.	●	○	●
Liu <i>et al.</i> [25]	2023	Deep learning on Tor-based WF attacks and defenses.	●	●	○
Chao <i>et al.</i> [28]	2024	Vulnerabilities, Attacks, Defenses, and Formalization in Anonymity Networks.	●	●	○
Our work	2025	Comprehensive analysis of strategies, challenges, and approaches in Tor-based WF attacks, defenses, and associated datasets.	●	●	●

●: All previous studies included. ●: Some previous studies included. ○: No previous studies included.

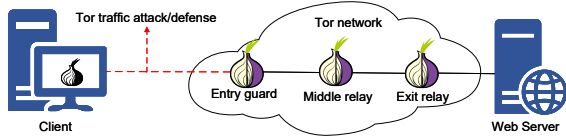


Fig. 2: Overview of the WF attack and defense location in Tor architecture.

than providing a systematic analysis of attack and defense strategies themselves. Additionally, it lacks deeper insights into the characteristics of WF attacks and defenses, and does not provide an extensive discussion on ongoing challenges, future directions, or evolutionary trends. Chao *et al.* [28] presented a comprehensive survey of anonymity networks with a particular focus on Tor, I2P, and Freenet. This paper systematically categorizes the vulnerabilities, attack strategies, and defense mechanisms that shape the ongoing battle between anonymity systems and adversaries. However, the survey primarily emphasizes protocol-level vulnerabilities, attacks, and defenses. While the authors briefly discuss some WF attack and defense studies, WF security is not the central focus. Consequently, the paper provides limited insights into WF attacks and defenses, particularly their broader implications for Tor between the client and the Tor entry node. To summarize, there is an urgent need to provide a comprehensive overview of the latest advancements in WF attack techniques or defensive mechanisms for advancing future research.

III. BACKGROUND

This section provides an overview of the Tor architecture, introduces the majority of Tor traffic features, and summarizes WF attack and defense techniques. Section III-A compares the existing representative anonymity techniques and clarifies the Tor architecture and why the Tor network could achieve user anonymity. Section III-B reviews the primary traffic features used in Tor traffic analysis. Section III-C describes the mechanics of WF attacks against Tor, and Section III-D surveys WF defense mechanisms.

A. Tor Architecture

Tor routes traffic through a series of randomly selected intermediary nodes called relays [1], [51], [52]. As shown in

Fig. 2, when a client initiates a connection, it selects three relays to form a **circuit**: an entry guard, a middle relay, and an exit relay. The client typically uses the circuit for approximately ten minutes before establishing a new one. The entry guard is the only relay that knows the client's IP address. The middle relay helps obscure the path between the entry and exit points, and the exit relay is responsible for forwarding the traffic to the final destination. Each relay in the circuit only knows its immediate predecessor and successor, ensuring that no single relay has complete knowledge of both the source and destination, thus preserving user anonymity. When a user (client) initiates a connection through the Tor network using the Tor Browser, the traffic is first encrypted multiple times using a process called onion routing. This layered encryption ensures that no single relay can simultaneously identify both the origin and destination of the communication, reinforcing Tor's core principle of anonymity.

Tor applies a layered encapsulation and segmentation process to ensure secure and reliable network packet delivery. Specifically, Fig. 3 illustrates the data representation across different layers of transport within the Tor network. At the application layer, Tor encapsulates data into fixed-size 512- or 514-byte cells, which serve as the fundamental unit of communication. These cells are subsequently encrypted and aggregated into TLS records, with each record typically containing several complete Tor cells. As TLS records traverse the network stack, they are further segmented into multiple TCP packets, constrained by the maximum segment size (MSS) of 1448 bytes for Ethernet. In the Tor network, each circuit can carry multiple streams, with each stream corresponding to a separate TCP connection. To manage flow control, Tor employs *SENDME* cells: one is sent for every 50 incoming cells per stream and one for every 100 incoming cells per circuit. In this way, Tor can produce traffic patterns that reflect the substantial data volumes generated by modern websites [11], [29].

As shown in TABLE II, Tor traffic protocol (Onion Routing) differs from traditional Hypertext Transfer Protocol Secure (HTTPS) protocols by transmitting data in fixed-size 512- or 514-byte cells, which helps obscure traffic patterns and resist analysis. HTTPS is a secure communication protocol that combines HTTP with Transport Layer Security (TLS). It

TABLE II: The comparison of representative encrypted traffic protocols.

Feature/Layer	HTTPS	VPN	QUIC	Tor (Onion Routing)
Encryption Layer	Application layer (TLS)	Network/transport layer (IPsec, OpenVPN, WireGuard)	Transport layer (over UDP)	Application layer (TLS) + multiple network layers
Encryption Scope	Only app data (e.g., web content)	Entire network traffic	App data with transport layer	Multi-layered encryption of full paths
IP Address Hidden?	✗ No	✓ Yes from websites ✗ No from VPN provider	✗ No	✓ Yes
Anonymity	✗ No	Partial (VPN sees identity and activity)	✗ No	✓ Yes
Traffic Obfuscation	✗ No	Depends on protocol used	✓ Yes	✓ Yes
DNS Leakage Protection	✗ No	✓ Yes, if DNS goes through VPN	✗ No	✓ Yes, DNS requests resolved within Tor
Resistant to Censorship?	Partially (depends on SNI and DoH)	Depends on obfuscation layer used	✗ No	✓ Strong resistance
Packet Size	~1,300 bytes (TCP + TLS payload)	1,400–1,500 bytes (depends on protocol)	1,200–1,400 bytes (UDP + QUIC)	Fixed-size (512- or 514-byte cells)
Performance	Fast	Moderate latency increase	Low latency	High latency due to relays

SNI: Server Name Indication; DoH: DNS over HTTPS.

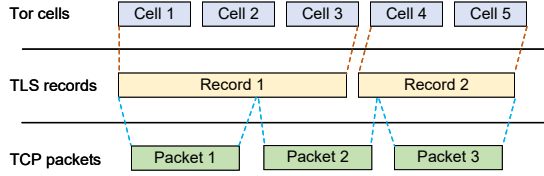


Fig. 3: Data representation between different layers of data transport.

encrypts data between a client (e.g., web browser) and a server, ensuring confidentiality, integrity, and authentication. Unlike HTTPS, QUIC is a protocol that encrypts both application data and transport-layer headers over the User Datagram Protocol (UDP). This protocol modifies the traditional TLS handshake to fit within its streamlined architecture. However, neither HTTPS nor QUIC conceals the client's IP address, packet size, or routing path, making them less effective against anonymity threats compared to Virtual Private Networks (VPNs) and the Tor network. In contrast, VPN encrypts a user's internet traffic and routes it through a remote server, masking the user's IP address and enabling access to geo-restricted content while maintaining relatively fast speeds. However, since the VPN provider can potentially monitor a user's activity, it is essential to choose a trustworthy service.

B. Tor Traffic Features

In WF attacks and defenses, the selection and extraction of traffic features are critical to classification accuracy and defense efficiency. Low-level features, such as the sequence of packet lengths and directions, capture fine-grained characteristics of a site's loading process. In contrast, higher-level features (i.e., coarse features), such as total upload/download bandwidth or the timespan of a trace, summarize traffic patterns in a more compact form. Attackers often combine multiple feature types to improve robustness against noise and varying network conditions. Correspondingly, defenders usually target these features by obfuscating or perturbing them. As shown in TABLE III, the following section presents a detailed

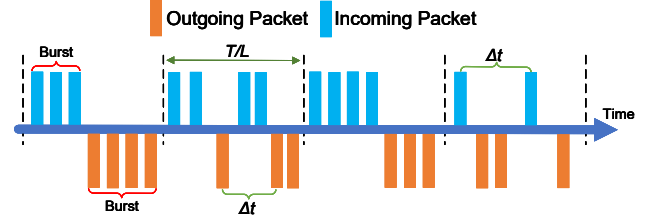


Fig. 4: Traffic timing trace with direction, bursts, and ordering. description of Tor traffic features and their applications in WF attack and defense research.

1) *Traffic Timing*: Timing features capture temporal patterns in network traffic that can reveal identifying characteristics of a website, even when the content is encrypted. As shown in Fig. 4, one commonly used metric is packet inter-arrival time [37], which measures the time interval Δt between consecutive packets or groups of packets. Similarly, the time difference between traffic traces and the round-trip time (RTT) provides insights into latency and responsiveness in the communication path [62], [65]. The time gap [62] reflects idle periods within a trace, often corresponding to content loading phases or user interactions. Other measures, such as the ratio of transmission time to total trace time [9], [58], [63], [64], summarize the distribution of active and idle periods within a session. Overall, these timing features help adversaries or analysts distinguish between websites by exploiting subtle but consistent timing signatures.

Timing features present a challenging surface to protect. Unlike packet sizes or directions, which can be easily padded or reshaped, time-based features are inherently tied to user interaction, network conditions, and server response behaviors. Defenses such as adaptive padding [16], [77], timing obfuscation [53], [72], and randomized delays [78] have been proposed to mitigate timing-based attacks. However, these approaches often introduce latency or degrade performance. As a result, incorporating traffic timing awareness in both attacks and defenses is now essential for evaluating the real-

TABLE III: The summary of important features and how to use the features for WF attack and defense techniques.

Features	Feature Description	How to use it for WF attacks?	How to use it for WF defenses?
Traffic timing	Tor cells are sent and received over time between the client and entry node. Even though the traffic is encrypted, its timing patterns are strongly influenced by a website's structure, resources, and delivery behavior.	Packet inter-arrival time [37]. Time gap [62]. Transmission time/Total trace time [9], [58], [63], [64]. The delay between the first outgoing packet and the first incoming packet [62], [65].	Random/adaptive dummy packets to vary the timing patterns [53]–[55]. Traffic regularization to normalize the timing patterns [56]–[58]. Adversarial perturbation to disturb the timing patterns [59]–[61].
Packet length	Different websites produce distinctive sequences of packet lengths because of variations in their webpage payloads.	Total number of packets [9], [66]. Number of outgoing packets [35]. Number of incoming packets [35]. The total length in bytes of the packet [37], [66].	In real-world scenarios, attackers cannot capture the total traffic length since Tor's start and end cells are unknown. So, WF defense studies exclude this feature.
Cell size	Websites differ in total packet/cell size due to varying content.	Packet/cell size. Total packet/cell size [7], [9], [35].	Excluded from WF defenses because of Tor's fixed-size cells.
Packet ordering	Different websites exhibit distinct request–response patterns, such as the alternation of client requests and server responses, the clustering of consecutive packets in one direction, or the timing of specific sequences.	Packet concentration and ordering features list [7], [35]. Average of the outgoing packet ordering list [64]. Standard deviation of the outgoing packet ordering list [35]. Time sequence/time-ordered manner [33], [70]. Number of incoming and outgoing cells before and after the candidate cell [37].	Add randomness to the traffic stream, such as injecting dummy packets [55], [67], [68] and introducing fake delays [58], [69]. Reordering packets by Traffic morphing [19], [22], [23] and creating a single "supersequence" of packets [9].
Traffic Bursts	Traffic bursts are short, high-rate sequences of packets (often dominated by either uploads or downloads) separated by intervals of relatively low activity, arising as a browser fetches different resources such as images, videos, and scripts at various times.	The frequency distribution of the packets or cells for a time interval [8], [9], [33], [71]. A sequence of outgoing packets, which is triggered by one incoming packet [11], [36].	Obfuscate characteristic packet sequence [11], [16], [72]. Create a consistent flow [57], [58], [73]. Obfuscate traffic flow classification [19], [23], [46], [74].
Direction	The incoming/outgoing (upload/download) traffic flow.	Assigning +1 to outgoing packets and -1 to incoming packets [12], [33], [39], [40].	Defense models cannot change the packet direction.
Coarse features	In website fingerprinting, coarse features are high-level traffic statistics that reflect distinct patterns in website content and loading behavior, such as bandwidth consumption and time-window-based counting.	Total transmission time or total timespan of a trace [75]. Total per-direction bandwidth [58]. Traffic "burstiness" and burst bandwidth [76]. Time-window-based counting [29].	Adjusting transmission rates or volumes to flatten bandwidth and hide burstiness. Morphing traffic distributions (<i>e.g.</i> , timespan, bandwidth) to mimic other sites.

world feasibility and robustness of WF techniques.

2) *Traffic Length*: Traffic length features capture the overall size or volume of a traffic trace, typically measured in terms of the number of packets, cells, or bytes exchanged during a browsing session [9], [36]. In WF attacks, traffic length is often highly informative because different websites require varying amounts of data to load. For instance, a media-rich website generally produces longer traces than a minimalist blog. Adversaries exploit this variability by treating traffic length as a coarse but effective discriminative feature, particularly in closed-world settings where the set of candidate websites is fixed and well-known. Even under encryption or anonymity systems, the total session length can still reveal strong clues about the visited website. However, in real-world scenarios, attackers face challenges in identifying the precise start and end points of a webpage load. To address this, some studies [62], [64], [65] approximate traffic length using time gap-based measurements. As shown in Fig. 4, T/L represents the fixed time gap or traffic length for captured Tor traffic flow. While this approach helps mitigate alignment issues, it can reduce accuracy when two different webpages produce traffic traces with similar lengths and time gaps.

From a defense perspective, attackers cannot precisely capture the total traffic length of a webpage because the start and end Tor cells are unknown. Therefore, WF defense studies rarely consider the traffic length feature.

3) *Cell Size*: In Tor, traffic is encapsulated into fixed-size units called cells (typically 512 or 514 bytes), so attackers cannot directly infer information from individual cells [37]. In early studies, researchers captured Tor traffic in a closed-world setting, where the total cell count and packet length for each webpage were known. Thus, cell size is widely used in traditional ML-based WF attacks [7], [9], [35], [79], [80]. However, in real-world scenarios, attackers cannot precisely measure the total traffic length because the start and end Tor cells are unknown. Consequently, recent website fingerprinting studies seldom rely on cell size for website classification.

4) *Packet Ordering*: Packet ordering features refer to the sequence in which packets or cells are transmitted and received during a browsing session, capturing the precise arrangement of request-response interactions between a client and a server [7], [35]. In WF attacks, attackers can analyze these ordered sequences to uncover consistent patterns that are unique to specific websites. For instance, certain pages may

always begin with a small outgoing request followed by a predictable series of large incoming responses, or they may load specific resources in a particular order. By modeling the order of packets, along with their direction and timing, ML algorithms can effectively fingerprint websites from encrypted Traffic. This makes packet ordering a subtle yet powerful indicator of web activity, especially when used in conjunction with other features, such as inter-packet timing and burst behavior [35], [64].

From a defensive standpoint, disrupting recognizable packet ordering is a complex task, as it requires altering the natural flow of web traffic without degrading usability or breaking functionality. Defenses that aim to obscure packet order often rely on techniques such as injecting dummy packets [55], [67], [68], randomized packet delays [58], [69], or traffic shaping [19], [22], [23], which attempt to introduce noise into the observed sequence of packets. However, these approaches come with significant trade-offs, including added latency, reduced performance, and the risk of interfering with application-layer protocols. Moreover, attackers can sometimes adapt to these defenses by leveraging probabilistic models that tolerate small variations in ordering. As a result, while packet ordering is a less frequently discussed feature compared to direction or timing, it plays a critical role in advanced WF attacks and must be carefully considered in the design of robust and practical defenses.

5) *Traffic Bursts*: Traffic bursts are defined as sequences of consecutive packets in the same direction, either incoming or outgoing, grouped between directional switches. In WF attacks, burst patterns capture higher-level structural information about how websites load resources, reflecting how browsers and servers exchange data in grouped transactions. For example, a user request (outgoing burst) might be followed by multiple server responses (incoming burst), forming a distinct burst sequence unique to the website [71]. These burst patterns, characterized by burst size, burst count, and burst order, are often stable across visits and provide a rich feature set for ML and DL models. SOTA attacks leverage bursts as highly discriminative features for classifying encrypted traces, prompting most WF defenses to focus on obscuring them.

Defending against burst-based fingerprinting requires strategies that disrupt the predictability of burst structure without significantly impacting user experience. Techniques like burst molding, dummy burst insertion, or splitting and merging of bursts have been proposed to obscure these patterns. Defenses such as Walkie-Talkie [45] and WTF-PAD [16] specifically target burst behavior by reshaping traffic to mimic the burst patterns of multiple websites or injecting randomness into burst timing and size. However, these approaches often introduce challenges such as increased latency, bandwidth overhead, or implementation complexity. Additionally, attackers demonstrated that even partially obscured burst patterns can leak identifiable information when combined with other features. Therefore, traffic bursts represent a critical feature class in both WF attacks and defenses, requiring sophisticated and adaptive countermeasures to mitigate their fingerprinting potential effectively.

6) *Traffic Direction*: Traffic direction features refer to the sequence and pattern of incoming and outgoing packets or cells during a user's interaction with a website. In WF attacks, the direction of traffic, represented as sequences of +1 (outgoing) and -1 (incoming), forms one of the most distinguishing features for identifying websites [12], [33], [39], [40].

Different web pages load resources in unique patterns: some initiate many small outgoing requests followed by large incoming responses (e.g., media-heavy sites), while others might produce more balanced or irregular exchanges. These directional patterns can be captured by ML models to fingerprint specific websites, even when encryption hides the actual content. Directional sequences are compelling when combined with other features, such as timing and packet counts, as they provide structural insight into how data flows during page loads. However, since packet direction cannot be altered, defense models typically exclude this feature.

7) *Coarse-grained Features*: Coarse-grained features in website fingerprinting refer to high-level traffic statistics that enable website identification without requiring decryption. They exploit differences in website loading behaviors, producing unique traffic patterns that persist even under encryption and fine-grained defenses. Per-direction bandwidth reveals the ratio of incoming to outgoing data. For example, a video streaming site will have a massive incoming bandwidth and a relatively small outgoing bandwidth, while a file-upload site will exhibit the opposite. Traffic burstiness refers to the tendency of data transfer to occur in bursts rather than at a constant rate. Even if some defense models try to obfuscate burst patterns in Tor traffic, they often fail to address traffic burstiness and burst bandwidth. These coarse-grained features can still leak significant information and allow for website fingerprinting [58].

Research on WF attacks targeting coarse features remains limited, and no dedicated defense methods have been proposed to date. However, based on existing studies, two possible defense approaches are: (i) Adjusting transmission rates or volumes to flatten bandwidth and hide burstiness. This involves smoothing out the traffic flow to eliminate the tell-tale peaks and valleys that reveal a website's fingerprint; (ii) Morphing traffic distributions (e.g., timespan, bandwidth) to mimic other websites. By actively changing a website's traffic pattern to match that of another, a defender can make it indistinguishable from others within a specific group.

C. WF Attacks

Website fingerprinting (WF) [11], [27], [35], [36], [81], also known as a WF attack, is a type of traffic analysis technique used to infer which websites a user is visiting, even when the traffic is encrypted and anonymized through tools like VPNs or the Tor network. We compare the current representative encrypted traffic protocols in TABLE II. By analyzing patterns in network metadata, such as packet sizes, timing, and sequences, attackers can bypass encryption (e.g., HTTPS, VPNs, or Tor) and infer browsing activity. These attacks pose significant risks to user anonymity, particularly in environments with heavy surveillance or censorship. Therefore, WF attacks pose

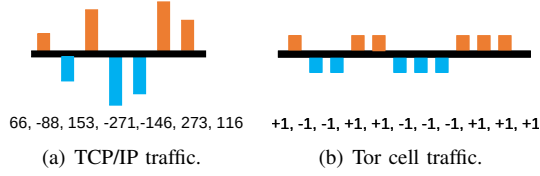


Fig. 5: An overview of the two primarily utilized forms of traffic data in WF studies.

a significant threat to user anonymity on the Tor network. Adversaries can exploit metadata patterns, such as packet sizes, timing, and sequences, to infer the websites a user visits. These attacks undermine Tor’s privacy guarantees, particularly for users in censored or surveilled environments.

1) *Traditional Machine Learning Methods:* Wagner *et al.* [82] designed and successfully implemented the first website fingerprinting and conducted preliminary experiments on a small number of websites. The concept of the WF attack was first proposed by Hintz [83] in 2002. They used web object size as fingerprints and verified the validity of WF attacks on the encrypted web proxy SafeWeb in small-scale experiments. Yan and Kaur [14] identify visited websites by analyzing TCP/IP header data, even when content is encrypted. The study focuses on a limited set of features by conducting an exhaustive analysis of TCP/IP features across diverse communication scenarios. The goal is to understand the full extent of website fingerprintability and identify previously unknown informative features. The early WF attacks are mainly performed against HTTPS [11], encrypted web proxies [58], [84], OpenSSH [66], [85]–[87], OpenVPN [86]. In the case of HTTPS, OpenSSH, and OpenVPN, researchers have gathered TCP/IP data packets generated when visiting HTTPS websites. The traffic is then transformed into a vector based on the order and direction of the data packets, as shown in Fig. 5(a). Until 2009, Herrmann *et al.* [8] performed the attack on the Tor network for the first time, which was based on a Naïve Bayes classifier with the features from the normalized frequency distribution of packet size. However, only 2.96% accuracy is obtained. Since then, WF attacks against Tor have become a research hotspot. And machine learning was applied to improve accuracy, including support vector machines (SVMs) [4], [7], [35], hidden markov models (HMM) [7], [35], and k-nearest neighbors (k-NN) [9]. We briefly introduce traditional ML methods as follows.

- Naïve Bayes [88] is a supervised ML algorithm based on Bayes’ theorem and the assumption of conditional independence among features. It calculates the posterior probability of each class given the observed features and selects the class with the highest probability, making it suitable for both binary and multiclass classification tasks. Due to its simplicity, scalability, and effectiveness in handling high-dimensional data, Naïve Bayes is widely used as a baseline model in ML.
- SVMs [89] are a class of supervised learning algorithms widely used for classification and regression tasks in ML. It works by finding the optimal hyperplane that best separates data points of different classes with the maximum margin. For non-linearly separable data, SVM

uses kernel functions, such as polynomial, Gaussian, and sigmoid, to transform the data into a higher-dimensional space where separation is possible. SVM is known for its effectiveness in high-dimensional spaces and its ability to generalize well with limited training data.

- k-NN [90] is a commonly used supervised learning method. It is a non-parametric, instance-based algorithm that makes no assumptions about the data distribution and uses the training data directly for prediction. Its simplicity and effectiveness allow it to identify the k closest samples in the training data and make predictions based on the majority label (for classification) or the average value (for regression) of these neighbors.
- Decision Tree [91] is another supervised learning algorithm used for both classification and regression tasks. It models decisions and their possible consequences in a tree-like structure, where each internal node represents a test on a feature, each branch corresponds to the outcome of the test, and each leaf node represents a final prediction or decision. ID3, C4.5, and CART are commonly used algorithms for constructing decision trees.
- Random Forest [92] builds upon the concept of decision trees by creating a “forest” of multiple trees. Random Forest achieves improved accuracy and robustness by aggregating the predictions from many diverse trees. This approach helps overcome the common problem of overfitting that individual decision trees often face.

2) *Deep Learning Methods:* Recently, a new class of WF attacks leveraging deep learning (DL) has emerged, aiming to enhance the accuracy, effectiveness, and robustness of Tor traffic analysis [10], [12], [87], [93]–[98]. These DL-based approaches have consistently achieved accuracies exceeding 90%, establishing themselves as SOTA WF attacks and serving as benchmarks for evaluating subsequent attacks and defenses. By exploiting structured input features and harnessing the powerful representation learning capabilities of DL architectures, these methods can automatically extract complex patterns from encrypted traffic, leading to superior classification performance. A brief overview of the representative DL models used in WF attacks is presented below.

- Convolutional Neural Networks (CNNs) [99] are a class of DL models particularly effective for automatically extracting hierarchical spatial and temporal patterns without requiring handcrafted features. They use convolutional layers to automatically learn and extract local features by applying small filters across the input, capturing spatial or temporal patterns. Pooling layers then reduce dimensionality while preserving essential information, and fully connected layers integrate the extracted features for classification or regression tasks. CNNs excel at detecting hierarchical patterns, making them widely used in fields such as computer vision, natural language processing, and traffic analysis.
- Long Short-Term Memory (LSTM) [100] is a type of recurrent neural network (RNN) designed to capture long-range dependencies in sequential data effectively. By incorporating memory cells and gating mechanisms

(input, output, and forget gates), LSTMs can selectively retain or discard information over time, effectively mitigating the vanishing and exploding gradient problems common in traditional RNNs. This capability makes them particularly effective for modeling temporal patterns and context in tasks such as natural language processing, speech recognition, and traffic analysis.

- Generative Adversarial Networks (GANs) [101] are a class of DL models consisting of a generator and a discriminator that are trained simultaneously in a competitive process. The generator creates synthetic data intended to resemble real data, while the discriminator attempts to distinguish between real and generated samples. Through this adversarial training, the generator learns to produce highly realistic outputs. GANs are widely used for tasks such as image synthesis, data augmentation, and anomaly detection.
- Transformers [102] are a type of DL architecture designed to handle sequential data by relying entirely on an attention mechanism, rather than recurrence or convolution. The attention mechanism enables the model to weigh the importance of different elements in a sequence, capturing long-range dependencies efficiently. Transformers are highly parallelizable and have achieved SOTA results in natural language processing, computer vision, and other sequence modeling tasks.
- Graph Neural Networks (GNNs) [103] are a class of neural networks designed to work directly with graph-structured data. They learn to represent nodes, edges, and entire graphs by aggregating and transforming information from a node's neighbors. This enables GNNs to capture complex relationships and patterns in data where connections are crucial, such as social networks, molecules, and transportation systems. GNNs have been successfully applied in various domains, including recommendation systems, chemistry, and network analysis.

3) *Evaluation metrics*: Evaluation metrics are essential for assessing the efficacy of WF attack models. The following content summarizes the key metrics used to measure WF attack performance.

Accuracy / Attack success rate (ASR): In closed-world scenarios, accuracy is the most commonly used metric for evaluating defense performance. It represents the percentage of traffic sequences that a WF attacker correctly classifies [72]. The definition of accuracy is shown below:

$$Accuracy(ASR) = \frac{N_{acc}}{N_{all}} \quad (1)$$

N_{acc} is the number of traffic sequences that the attacker correctly classifies; N_{all} is the total number of traffic sequences tested.

TPR , FPR [46], [72], [104]: The TPR and FPR are commonly used for evaluation in the open-world setting. The TPR demonstrates the percentage of correctly labeled monitored traces over the total number of monitored traces, and the FPR describes the percentage of the non-monitored traces that are misclassified as monitored ones.

$$TPR = \frac{TP}{TP + FN} \quad (2)$$

$$FPR = \frac{FP}{FP + TN} \quad (3)$$

TP represents the number of samples correctly classified in the set of monitored sites. TN represents the number of samples correctly classified in the set of unmonitored sites. FP denotes the number of samples misclassified as belonging to the monitored sites. FN indicates the number of samples misclassified as belonging to the set of unmonitored sites.

F1-score and Precision-Recall curves: In open-world scenarios, the F1-score and Precision-Recall curves are commonly used to directly represent performance. The F1-score can be viewed as a weighted average of precision and recall, where a lower F1-score indicates weaker classification performance. The Precision curve illustrates the trade-off between precision and recall; a lower curve position reflects reduced classification accuracy, indicating a more effective defense.

$$F1 = \frac{2TP}{2TP + FN + FP} \quad (4)$$

$$Precision = \frac{TP}{TP + FP} \quad (5)$$

$$Recall = \frac{TP}{TP + FN} \quad (6)$$

D. WF Defenses

The development of effective defense mechanisms against website fingerprinting has become a critical area of research in online privacy. As introduced in the previous section, WF is a traffic analysis attack that enables an adversary to infer which website a user is visiting by analyzing patterns in encrypted traffic. Despite the use of strong encryption protocols, features such as packet sizes, timing, direction, and burst patterns can still reveal identifiable characteristics of web traffic. When Hintz [83] raised the first WF attack, they also put forward the methods to defend against the WF attack, including adding extra noise data to the traffic, reducing the number of files transferred, and transferring everything in one connection.

1) *WF defense categories*: Website fingerprinting defenses aim to obscure traffic patterns that attackers rely on to infer which websites users are visiting, even when the data itself is encrypted. Since early research focused on traditional single-proxy systems, similar to defense research, such as link padding [105], traffic morphing [17], [106], and HTTPoS [107]. In 2011, Panchenko et al. [4] made the first defense against WF on Tor. They applied a camouflage strategy to confuse the attack classifiers, which attempted to load another page in parallel to obfuscate the page. Early defenses in Tor primarily focused on padding-based approaches, which insert dummy packets to obscure packet size and burst features. While effective in reducing information leakage, padding often incurs high bandwidth overhead. Later, traffic shaping and regulation techniques were introduced to create more uniform traffic flows by delaying, batching, or smoothing

packet transmissions. These methods improve robustness but may add significant latency.

More recent defenses explore traffic morphing strategies, which transform traffic distributions to resemble those of other websites [19], [24]. Additionally, they employ adversarial perturbation techniques inspired by adversarial machine learning [46]–[48], which subtly modify traffic patterns to mislead classifiers. Despite these advances, designing an effective WF defense remains challenging due to the trade-off between privacy, usability, and system performance. Many defenses successfully disrupt fine-grained features (*e.g.*, packet order or timing sequences) but remain vulnerable to coarse-grained features such as total timespan, bandwidth ratios, or burst-level statistics. This motivates ongoing research into more comprehensive and adaptive defense mechanisms that can protect against increasingly sophisticated WF attacks. A detailed comparison of the strengths and limitations of representative WF defense strategies will be presented in Section V-B. A brief overview of the WF defense categories in this paper is presented below:

- Adaptive Padding [53] is a WF defense technique that seeks to hide traffic patterns by inserting dummy packets in a context-aware manner. Unlike fixed padding schemes, adaptive padding selectively adds cover traffic based on observed packet sequences, aiming to obscure identifiable bursts without introducing excessive overhead. By adaptively adjusting padding according to traffic dynamics, adaptive padding reduces the distinctiveness of packet direction and timing features while maintaining lower bandwidth and latency costs compared to constant-rate padding methods.
- Traffic regularization [108] reshapes network traffic to create a more uniform and predictable flow. Instead of allowing natural bursty patterns, packets are delayed, batched, or transmitted at fixed intervals, making it harder for attackers to exploit timing or burst features. By smoothing out traffic variations, traffic regularization conceals fine-grained patterns, though it often introduces additional latency and may reduce network efficiency.
- Traffic morphing [9] is a WF defense technique that transforms a website's traffic patterns to mimic those of another site. By modifying features such as packet sizes, burst lengths, or overall traffic distribution, morphing reduces the distinguishability of traffic traces. This approach aims to confuse classifiers by making different websites appear statistically similar, though it can introduce bandwidth overhead and may not scale well across diverse traffic types.
- Adversarial perturbation [109] introduces carefully crafted, small modifications to traffic patterns, such as slight delays, dummy packets, or reordering, that are imperceptible to users but mislead ML classifiers. The goal is to degrade the accuracy of WF attacks while keeping bandwidth and latency overhead relatively low. However, its effectiveness depends on the attack model, and adaptive adversaries may still retrain to counter it.

2) *Evaluation metrics*: Evaluation metrics are essential for assessing the performance of a defense mechanism. The following content summarizes the key metrics used for this purpose.

Accuracy: Accuracy for defense samples training is the same as standard classification accuracy.

$$Accuracy_{def} = \frac{N_{correct\ defense\ samples}}{N_{total\ defense\ samples}} \quad (7)$$

$N_{correct\ defense\ samples}$ is the number of defense samples the model still classifies correctly. $N_{total\ defense\ samples}$ denotes the total number of defense samples.

Defense success rate (DSR):

$$DSR = \frac{N_{misclassified\ defense\ samples}}{N_{total\ defense\ samples}} \quad (8)$$

$$DSR = 1 - Accuracy_{def} \quad (9)$$

$N_{misclassified\ defense\ samples}$ denotes the number of defense examples that cause the model to output the wrong label. $N_{total\ defense\ samples}$ denotes the total number of defense examples generated.

Bandwidth overhead / Data overhead [57], [72]: The data overhead $O(D)$ of defense method D on P is defined as the ratio of the total amount of dummy data to the total amount of real data. P denotes the original trace. P' denotes the trace after implementing some defense D .

$$O(D) = \frac{|P'| - |P|}{|P|} \quad (10)$$

$|P|$ denotes the total number of cells in the trace. $|P'|$ is the total number of cells in the defense trace.

Latency overhead / Delay Overhead [57], [72]: Adding dummy packets can increase transmission time. Delay overhead is the extra time caused by the defense divided by the original transmission time. The latency overhead $L(D)$ of defense method D on P is the extra time taken to transmit real packets, divided by the original transmission time.

$$L(D) = \frac{t_k - t_{|P|}}{t_{|P|}} \quad (11)$$

t_k denotes the time spent transmitting the defended trace P' . $t_{|P|}$ denotes the time spent transmitting the original trace P .

IV. DATASETS

In this section, we summarize representative public datasets to help researchers select suitable datasets for their studies. With the widespread application of ML and DL techniques in WF attacks and defenses, datasets play a crucial role in performance evaluation. TABLE IV presents the dataset sizes for both closed- and open-world settings, along with their features and whether they include multi-tab browsing.

A. Dataset Categories

A critical aspect of evaluating WF attacks and defenses is the dataset used for training and testing. These datasets are typically categorized into two main types: closed-world and open-world [9], [11], [94]. The choice of dataset has a significant impact on the attack model's assumptions and the practical relevance of evaluations.

1) *Closed-world dataset*: A closed-world dataset simulates a scenario where an adversary has a complete list of all possible websites a user might visit. The client is assumed to visit a limited set of websites referred to as monitored websites [7], [36], [37]. The dataset consists of a finite, pre-defined set of websites, and the machine learning model is trained to classify a given traffic trace into one of these specific sites. However, this model assumes the adversary has complete knowledge of the user's browsing space. While this assumption is not realistic for the general internet, it serves as a valuable benchmark for evaluating the maximum potential accuracy of a fingerprinting attack under ideal conditions. Researchers use closed-world datasets to test the effectiveness of new algorithms and features without the added complexity of unknown traffic [95].

2) *Open-world dataset*: An open-world dataset provides a more realistic and challenging scenario. In an open-world setting, the dataset is divided into two categories: a small set of "monitored" or "sensitive" websites and a much larger set of "unmonitored" or "unknown" websites [9]. The attacker's goal is to accurately identify traffic from the monitored sites while correctly classifying all other traffic as "unknown" [35].

The open-world dataset is a more accurate representation of a real-life scenario, as an adversary would never have prior knowledge of every possible website a user could visit. Evaluating attacks and defense on open-world datasets is crucial for understanding their practical effectiveness, as it accounts for the challenge of distinguishing between targeted sites and the vast, diverse landscape of the internet. The primary challenge in this setting is avoiding the base-rate fallacy, where a classifier with a high true positive rate might still produce an overwhelming number of false positives due to the sheer volume of unmonitored traffic.

B. Public Dataset

Cai *et al.* [7] collected traffic traces from the Alexa [111] Top 1,000 websites in 2012, ultimately retaining the top 800 cleaned URLs after removing failed pages and resolving redirects. For each URL, they gathered either 20 or 40 traces. Two closed-world Tor traffic datasets were then constructed: (1) one with 100 websites and 20 traces per site, where all packet size information was removed; and (2) another with 100 websites and 40 traces per site, where both packet size and packet direction information were removed. However, no open-world dataset was included. Later, Wang and Goldberg [35] extended this line of work by adding 860 non-monitored traffic instances and 40 monitored instances from Alexa's Top 1,000 sites to evaluate their model. To collect Tor traffic data, they disabled browser caching, prevented caching to disk, and cleared the browser cache each time

the Tor Browser was restarted. However, multi-tab browsing was still not considered, and the collected dataset has limited diversity. Subsequently, Wang *et al.* [9] released a new dataset in 2014 (named Wang14) that included both closed- and open-world scenarios. They collected 90 instances for each of 100 monitored (sensitive) pages, along with one instance each for 5,000 non-monitored pages from Alexa's Top 10,000, excluding any sites already included in the monitored set by domain name. In their data collection, webpage loading was performed with regular circuit resets, disabled caching, and time gaps between multiple loads of the same page.

Previous Tor traffic datasets for WF attacks and defenses often overlooked the diversity of sub-pages (*e.g.*, articles, profiles) by focusing primarily on single pages and operating at too small a scale to reflect the real Internet. To address this limitation, Panchenko *et al.* [11] constructed larger and more diverse datasets. They discarded duplicate entries but deliberately included different web pages from the same website when available. Query parameters such as session identifiers were removed, as they would prevent reproducible retrievals. Furthermore, all URLs pointing to pure advertisement services were excluded, since these are clearly not the user-intended target pages. By combining multiple source datasets [9], [35], the authors produced two large-scale datasets: one containing 34,580 unique websites (118,884 unique web pages) and another containing 65,409 unique websites (211,148 unique web pages). However, Hayes *et al.* [36] later showed that the feature set of these datasets, which relies heavily on packet order and counts, is highly vulnerable to even simple WF defenses. To conduct their study, they collected two subsets of monitored web pages from Alexa's top 20,000 websites: (1) 100 instances each from the top 55 Alexa-ranked web pages and (2) 80 instances each from 30 popular Tor hidden services (HS). A Tor hidden service is a website hosted behind a Tor client's Onion Proxy, which acts as the interface between the application and the network.

Previously collected datasets were relatively limited in size, both in terms of classes (*i.e.*, the number of unique websites) and instances (*i.e.*, the number of traffic traces per website). To address this limitation, Rimmer *et al.* [94] constructed a closed-world dataset consisting of 900 websites, with 2,500 traffic traces collected for each. They also built an open-world dataset comprising 400,000 unmonitored websites and 200 monitored websites selected from Alexa's top 400,000 list. To account for temporal variation, Oh *et al.* [112] collected 110 traffic instances for each of 100 websites across four separate batches from Alexa's top 100 websites. The batches were collected sequentially, with each subsequent batch starting one week after the previous one. Wang *et al.* [45] collected their dataset from long-standing pages, selecting 100 of the top sites as the non-sensitive set (after removing duplicates caused by different localizations) and using the next 10,000 pages from Alexa's top list as the sensitive set. Sirinam *et al.* [93] built a closed-world dataset of 95 websites with 1,000 traces per site from Alexa's top 100 list, as well as an open-world dataset containing 40,716 traffic traces.

In 2020, Gong and Wang [57] collected a dataset, referred to as DS-19, from the homepages of Alexa's top 100 websites.

TABLE IV: A summary of representative public datasets.

Dataset Name	Year	Dataset Size		Features	Included Multi-tab
		Closed World	Open World		
Wang14 [9]	2014	90×100	90×100 + 5,000×1	Traffic timing, Packet length, Traffic direction, Packet ordering, Traffic bursts	✓
CUMUL [11]	2016	211,148	118,884	Traffic timing, Cell size, Packet length, Traffic direction, Traffic ordering, Traffic bursts	✗
k-fingerprinting [36]	2016	100×55 HS: 80×30	NA	Traffic timing, Packet length, Traffic direction, Traffic ordering, Traffic bursts	✗
AWF [94]	2017	900×2500	400,000+200	Traffic direction, Traffic ordering, Traffic bursts	✗
Walkie-Talkie [45]	2017	100×100	9,900×1	Traffic timing, Packet length, Traffic direction, Traffic ordering, Traffic bursts	✗
DF [93]	2018	95×1000	40,716×1	Traffic timing, Cell size, Packet length, Traffic direction, Traffic ordering, Traffic bursts	✗
DS-19 [57]	2020	100×100	10,000×1	Traffic timing, Cell size, Packet length, Traffic direction, Traffic ordering, Traffic bursts	✗
Tik-Tok/W-T [12]	2019	100×1	10,000×1	Traffic timing, Packet length, Traffic direction, Traffic ordering, Traffic bursts	✗
ARES [110]	2023	100×1,000	100×1,000	Traffic timing, Traffic direction, Traffic ordering, Traffic bursts	✓
NetCLR [71]	2023	90×(100+20)	5,000×1	Traffic timing, Cell size, Packet length, Traffic direction, Traffic ordering, Traffic bursts	✗
TMWF [43]	2023	Chrome _S : 50×100 TBB _S : 50×100	Chrome _S : 5,000×1 Chrome _M : 10,000×1 TBB _S : 5,000×1 TBB _M : 10,000×1	Traffic timing, Traffic direction, Traffic ordering, Traffic bursts	✓

S: Single-tab dataset. *M*: Multi-tab dataset.

They gathered 100 traces of each homepage as the monitored set and 10,000 additional web pages as the unmonitored set, filtering out pages that failed to load (e.g., those inaccessible via Tor). Recognizing that an attacker cannot rely on training datasets containing only correct pairings, Rahman *et al.* [12] collected Tor traffic from 100 monitored websites and sampled 10,000 unmonitored websites from Alexa’s top 14,000 list. To accurately represent the browsing behavior of the Tor Browser bundle, the authors used a modified variant of the Tor Browser Crawler [113]. Similarly, Cadena *et al.* [114] built an open-world dataset consisting of 11,307 popular Alexa websites (excluding the top 100) along with 100 traces for each of Alexa’s top 100 websites. Wang [115] selected the top 100 Alexa pages as the sensitive set, visiting each 200 times, and the next 80,000 pages as the non-sensitive set, visiting each once. Gong *et al.* [116] created a new undefended dataset using WDefProxy [117], consisting of Tor traces from 100 monitored pages and 10,000 non-monitored pages, where each monitored page was visited 100 times and each non-monitored page once. To study the impact of concept drift in WF attacks, Bahramali *et al.* [71] collected up to 550 traces per visit for each of 225 non-onion websites in the closed-world setting. For the open-world setting, the authors gathered a single trace per website, selecting 10,000 unmonitored websites. The collected data was then divided into three subsets: (1) Drift90: a set of 90 monitored websites, each with at least 100 superior and 20 inferior traces; (2) Drift-Guard: the same 90 monitored

websites, further split into traces collected through 11 guard relays in Europe and 7 guard relays in North America; and (3) Drift5000: a set of 5,000 unmonitored websites.

To evaluate their proposed WF attack model in a multi-tab setting, Xu *et al.* [62] collected both single-tab and two-tab datasets. The single-tab dataset consists of three subsets: (i) 50 instances from each of 50 monitored web pages without background noise for training, (ii) another 50 instances from each of the same 50 monitored web pages without background noise for testing, and (iii) an open-world subset comprising instances from 2,500 unmonitored web pages. The two-tab dataset was constructed with five different time gaps: two, three, four, five, and six seconds. For each time gap, they collected traffic traces from 50 monitored web pages, with 50 instances per page. In addition, they collected 5,000 instances for use in dynamic split experiments. Guan *et al.* [118] collected 10,000 training traces and 1,000 testing traces from 50 websites. They generated real-world two-tab traces by visiting two sites 100 times each and combining the 100×100 possible pairs. They also constructed synthetic two- and three-tab traces by merging single-tab traces from the dataset in [45], selecting the 50 websites with the most packet traces (90 traces per site for training, the rest for testing). Each two-tab trace was formed by aligning two single-tab traces with an initial delay and interleaving packets by timestamp, ensuring every site appeared in both tab positions at least once, while the other tab was chosen randomly. To better reflect real-

world scenarios where multiple tabs may overlap, they further merged three traces per sample in the closed-world setting, with the additional two tabs randomly chosen from the 50 websites. This resulted in 13,500 training traces and 2,562 testing traces. Deng *et al.* [110] extended this line of work by collecting a five-tab dataset that comprises 10 dark web websites and 90 websites outside Alexa’s top 100 list. They also investigated scenarios with multiple Tor Browser versions, gathering five-tab traces from versions 10.0.15, 10.5, 11.0.3, 11.0.10, and 11.5. In addition, they built a five-tab dataset from 1,000 subpages of 100 websites.

Existing multi-tab datasets are typically created by manually synthesizing multi-tab traces from single-tab recordings. To enable a more realistic evaluation of multi-tab WF attacks, Jin *et al.* [43] collected both single-tab and 2-tab datasets for Chrome and Tor browsers (TBB). Their monitored set comprises 5,000 traces (50 monitored websites $\times$ 100 single-tab traces per site), and they collected an additional 5,000 non-monitored traces for each browser. For the open-world evaluation, they gathered over 10,000 2-tab trace samples for Chrome and Tor; these traces were generated by randomly selecting overlap ratios in the range [0.1, 0.5] so as to simulate diverse access and overlap scenarios. For traces with more than two tabs, the authors continued to rely on manual synthesis: multi-tab samples are created by concatenating single-tab traces using a controlled overlap ratio or by introducing randomized inter-tab delays to emulate real browsing timing. These synthesis strategies allow the authors to scale training sets, vary overlap characteristics, and evaluate model robustness across a range of realistic and adversarial multi-tab conditions.

V. THE STRATEGIES AND APPROACHES ON WF ATTACKS AND WF DEFENSES

In this section, we present a systematic classification framework that focuses on techniques where models serve as effective mechanisms for WF attacks or defenses. Precision and accuracy are central concerns for researchers in this field. According to the existing studies, WF attacks are categorized into two main groups: traditional machine learning models and deep learning models. To better reflect real-world scenarios, we also include an analysis of multi-tab WF attack approaches. On the defense side, we examine techniques aimed at mitigating WF attacks, including adaptive padding, traffic regularization, and adversarial perturbation methods. Lastly, we provide an overview of the datasets commonly used in WF attack and defense research.

A. Survey on WF attacks

Panchenko *et al.* introduced the first WF attack that accurately classified web pages on Tor in 2009. Since then, various techniques have been developed, broadly categorized into single-tab and multi-tab fingerprinting based on user browsing behavior [149], [150]. Multi-tab WF represents more realistic usage patterns, where users access multiple websites simultaneously in different tabs. However, it introduces added complexity, as traffic from multiple sites is interleaved, making it more difficult to distinguish and identify individual websites.

To help researchers choose appropriate models for WF attacks, this paper classifies existing WF attack methods into three categories: traditional ML-based, DL-based, and non-machine learning-based approaches. By incorporating traffic-level characteristics, such as direction, timing, and burst patterns, the proposed classification framework offers a more practical and application-oriented method for selecting suitable models. A summary of the current WF attack techniques and their applied features is presented in TABLE V.

Previous WF attacks on Tor have often relied on self-collected open-world datasets, which vary widely in terms of scale and environmental conditions, such as network settings, browser configurations, and webpage selections [11], [35], [94]. This inconsistency makes it difficult to compare the effectiveness of different WF methods directly. To address misleading claims of high recall or true positive rate (TPR) and low false positive rate (FPR) in existing WF attacks, Wang [151] proposed enhancing classifier performance through the use of precision optimizers based on confidence scoring, distance metrics, and ensemble learning. These optimizers prioritize precision over recall, which is better suited for open-world website fingerprinting (OWF), especially in large-scale surveillance settings where the base rate fallacy, the overestimation of performance caused by the rarity of target websites, can distort evaluation. Further, in 2020, Wang [115] highlighted that prior work often failed to properly account for base rates when calculating precision. To resolve this, the author introduced r-precision, a metric that explicitly incorporates the rarity of sensitive pages, offering a more realistic assessment of attack performance in open-world settings. To support more reliable and reproducible evaluations of WF attacks, this survey provides a comprehensive classification of WF models, dataset scales, and accuracy metrics, aiming to bridge the methodological gaps in prior research.

1) *Traditional ML-based WF Attack Models:* Traditional ML-based WF attacks represent some of the earliest and most extensively studied approaches in the field of traffic analysis [7], [8], [35], [58], [66]. These techniques rely on classical supervised learning algorithms to classify encrypted traffic traces based on statistical features extracted from network flows. Despite the encryption and anonymity provided by Tor, these ML models can often infer which websites users visit by analyzing observable traffic patterns such as packet sizes, directions, timings, and burst behaviors.

In a typical traditional ML-based WF attack, the attacker first collects labeled traffic traces from a set of monitored websites under controlled conditions. These traces extract a range of hand-crafted features, such as total packet count, inter-arrival times, or traffic burst lengths. These features are then used to train classifiers like SVMs [7], [35], [37], K-NN [9], [33], [37], [63], Random Forests [36], or Naïve Bayes [8], [37], [66], which learn to distinguish between traffic patterns corresponding to different websites. TABLE VI summarizes the existing traditional ML-based WF attacks, reporting their performance in both closed-world and open-world settings using practical evaluation datasets. If the dataset column is marked as *none*, it indicates that the authors used a self-collected dataset rather than a publicly available one.

TABLE V: Website Fingerprinting Attack Models and Applied Features

		Model	Traffic Timing	Cell Size	Packet Length	Direction	Packet Ordering	Bursts	Coarse features	Papers
Machine Learning	Machine Learning	Naïve Bayes				✓		✓		[8]
					✓	✓				[66]
			✓		✓	✓	✓			[37]
			✓			✓		✓	✓	[58]
		SVM		✓		✓	✓			[7], [35]
			✓		✓	✓	✓			[37]
			✓		✓	✓				[4]
		k-NN	✓	✓	✓	✓	✓	✓		[9]
			✓			✓		✓		[33]
				✓		✓	✓			[35]
			✓		✓	✓	✓			[37]
				✓		✓				[79]
			✓	✓	✓					[80]
		Decision Tree	✓			✓	✓	✓		[63]
		Random Forest	✓		✓	✓	✓	✓		[36]
		Hoeffding Tree	✓		✓	✓	✓	✓		[64]
		BalanceCascade-XGBoost	✓		✓	✓	✓	✓		[62]
			✓	✓	✓	✓		✓		[65]
	Deep Learning	CNN	✓			✓				[38], [96], [116], [119]–[121]
			✓			✓		✓		[12], [33], [39], [40]
			✓	✓		✓				[13], [94], [118]
			✓			✓				[42], [43], [122], [123]
				✓		✓				[124]
			✓		✓	✓		✓		[32]
						✓				[93], [110], [125]–[130]
						✓		✓		[131]
			✓			✓	✓		✓	[29]
		LSTM	✓			✓				[38]
			✓	✓		✓				[94]
		SADE	✓			✓				[38]
			✓	✓		✓				[94]
						✓				[10]
		Few-shot Learning	✓			✓				[38], [132], [133]
						✓				[128], [134]–[138]
					✓	✓				[95], [139]
						✓		✓		[140], [141]
		DNN	✓	✓	✓	✓				[112]
			✓			✓				[142]
		GAN	✓		✓	✓				[143]
						✓				[144]
		Transformer	✓			✓				[43]
			✓			✓	✓			[30]
						✓	✓			[70]
		GNN	✓		✓	✓				[31]
						✓			✓	[75]
		EDF-S and EDF-P	✓			✓				[145]
		LRCT				✓				[44]
		Enhanced Technique	✓		✓	✓				[146]
		NetCLR				✓		✓		[71]
		UNetFormer				✓		✓		[147]
		CIL	✓		✓	✓				[148]

TABLE VI: A summary of representative traditional ML-based WF models.

Research	Year	Architecture	Dataset	Closed World		Open World	
				Dataset Size	Acc (%)	Dataset Size	Acc (%)
Wang <i>et al.</i> [35]	2013	SVM and OSAD	CS-BUFLO [7]	100×20 100×40	91.00±2.00	860+40	91.00±6.00
DL-SVM [7]	2012	DL-SVM	CS-BUFLO [7]	100×20 100×40	90.00	NA	NA
K-NN [9]	2014	K-NN	Wang14 [9]	NA	NA	90×100	91.00±3.00
		Splitting algorithm and K-NN		NA	NA	90×100	97.80±0.60
Wang <i>et al.</i> [37]	2016	Splitting algorithm and Time-KNN	Wang14 [9]	NA	NA	90×100	97.00±2.00
		Splitting algorithm and SVM		NA	NA	90×100	90.00±3.00
k-fingerprinting [36]	2016	Random Forest	Wang14 [9]	550	91.50	6000	NA
AdaWFPA [64]	2019	Hoeffding Tree	Walkie-Talkie [45]	100×100	99.60	100×100	99.40
			Wang14 [9]	90×100	99.68	90×100	99.68

For other columns, *none* indicates that the corresponding information was not reported or could not be found.

Naïve Bayes-based WF Attacks: Naïve Bayes is one of the earliest and simplest machine learning algorithms applied to WF attacks. It is a probabilistic classifier based on Bayes' theorem with the assumption of feature independence, which makes it computationally efficient and easy to implement. In the context of WF, Naïve Bayes classifiers are trained on hand-crafted features extracted from encrypted traffic traces, such as total number of packets, packet size distributions, inter-arrival times, and traffic bursts [37]. These features are used to compute the likelihood of a traffic trace belonging to a particular website, and the classifier selects the website with the highest posterior probability. In 2006, Liberatore and Levine [66] first introduced packet lengths as a powerful WF feature for WF attacks. This paper first applied the Naïve Bayes classifier to map each packet sequence with unique packet lengths. Then, the authors used the Jaccard coefficient as a measurement of the distance between two packet sequences and achieved 96.65% accuracy on the OpenSSH dataset. In 2009, Herrmann *et al.* [8] performed the WF attack on the Tor network for the first time, which was based on a Naïve Bayes classifier with features from traffic flow direction and the normalized frequency distribution of observable IP packet size. In 2012, Dyer *et al.* [58] introduced a variable n-gram classifier that employs Naïve Bayes as its underlying machine learning algorithm. Their study found that coarse-grained features (Total transmission time, total per-direction bandwidth, and burst bandwidth) were particularly effective in enabling classifiers to distinguish traffic, even when it was protected by packet padding defenses. In summary, Naïve Bayes was shown to be effective in early WF studies despite its simplicity. However, the NB model assumes that all features

are independent, which is rarely true in traffic data. Moreover, Naïve Bayes tends to perform poorly in open-world settings, where distinguishing between monitored and non-monitored sites is significantly more challenging.

SVM-based WF Attacks: SVMs are another widely used method in early WF attacks due to their strong performance in high-dimensional classification tasks and their robustness to overfitting. SVMs are particularly attractive for WF because they perform well with limited training data and effectively handle high-dimensional feature spaces. In 2011, Panchenko *et al.* [4] were the first to apply SVM to WF attacks by defining features based solely on traffic volume, timing, and direction. Their approach significantly improved Tor website classification accuracy from 3% to 55%, and achieved a 73% recall rate in an open-world setting. Their approach is also the first successful WF attack under realistic open-world conditions. In 2012, Cai *et al.* [7] introduced an optimal string alignment distance (OSAD) to measure the website fingerprinting distance based on packet size for TCP packet traces. Then, the authors applied SVM to classify and recognize websites and web pages with 90% accuracy. Based on Cai's work, Wang and Goldberg [35] first introduced a method for representing Tor traffic as a one-dimensional array of size $[1 \times L]$, where each element indicates the direction of a Tor cell: +1 for outgoing and -1 for incoming packets. To standardize input lengths across all traffic traces, sequences were padded with zeros to a fixed length, commonly $L=5000$. This 1-D array representation has become a standard input format for many WF attack models. Despite their effectiveness, SVM-based WF attacks also face several challenges: (1) The accuracy of SVM-based WF attacks depends heavily on the selection of features. If the wrong features are chosen, or if features are noisy, the model may fail to classify traffic

correctly; (2) Performance can degrade in open-world settings due to the large and dynamic website changes; (3) The model may struggle with scalability as the dataset grows.

K-NN based WF Attacks: K-NN is a non-parametric, lazy learning algorithm used for classification and regression tasks. It works by finding the most similar instances (neighbors) to a given input sample and then assigning the class based on a majority vote from these neighbors. In the case of website fingerprinting, K-NN classifies network traffic based on the similarity of the features (such as packet sizes or packet ordering patterns) to the features of known traffic traces [37]. In 2013, Wang *et al.* [35] improved the K-NN algorithm by introducing time-based splitting and classification-based splitting to isolate traffic corresponding to single or multiple web pages. Their experiments demonstrated that time-based splitting could achieve up to 97.00% TPR for detecting single-page sequences, while classification-based methods further improved accuracy in cases of overlapping traffic. Later, Wang *et al.* [9] applied the K-NN classifier to a prominent feature set with weight adjustments in a realistic open-world scenario. Their results showed an 85.00% TPR with a FPR of 0.60%, outperforming previous work, which achieved 83.00% TPR with a FPR of 6.00%.

Previous studies [9], [35], [37] often neglect hyperlink transition information, as it introduces additional “noise” that complicates the classification of the original webpage. To address this issue, Zhuo *et al.* [79] proposed a WFP technique that integrates PHMM with K-NN, where PHMM is traditionally applied in bioinformatics for DNA sequence analysis. This method accounts for hyperlink transitions, making it more realistic and effective. The technique models websites by considering different webpages and their transitions, allowing for the identification of a sequence of page loads belonging to the same website. Many WF attacks assume websites do not change over time. In reality, websites are highly dynamic, leading to significant intra-class variability that reduces classification accuracy. Cui *et al.* [80] revisited key assumptions underlying WF attacks, which allow adversaries to infer users’ web activity despite encryption. By incorporating Markov models to capture temporal and probabilistic properties of web traffic, the paper highlights that WF attacks can remain effective under more realistic conditions. To distinguish between single-page and two-page traces, the authors employed a K-NN binary classifier. Their results show that the method can identify whether a trace contains one or two pages with an accuracy of 97.00%. Traditional K-NN classifiers typically rely on the labels of the nearest samples for classification, but the diversity of multi-tab traffic can lead to sample drift, degrading performance. To address this challenge, Zhao *et al.* [33] integrated proxy-based K-NN and sample-based K-NN to enhance multi-tab webpage identification. Their experiments showed that this hybrid model improved the multi-label recall metric by 88.60% compared to the state-of-the-art attacks.

Tree-based WF Attacks: Some researchers have explored Tree-based models to improve the accuracy of WF attacks. Tree-based models use decision trees [63] or ensembles of trees [62], [65] to learn from features extracted from network

traffic. These models recursively split the data based on feature thresholds, aiming to maximize the separation of traffic associated with different websites. In 2015, Kwon *et al.* [63] used C4.5 and CART decision trees to classify Tor hidden service circuits. The authors demonstrate that even without compromising the Tor network or engaging in active traffic manipulation, it is possible to infer which hidden service a Tor client is accessing. By leveraging decision trees and WF, adversaries can deanonymize users and services with high accuracy (>99.00%). To address the challenge of a large amount of noisy data, Hayes *et al.* [36] introduced k-fingerprinting, which leverages random decision forests to classify encrypted or anonymized web traffic. The method transforms packet sequences into distinctive “fingerprints” by mapping them to the leaf node identifiers produced by a trained random forest. It achieves impressive results, with TPR of up to 85% and FPR as low as 0.02% when identifying monitored hidden services.

The accuracy of WF attack models drops drastically when tested on captured traffic traces of websites due to concept drift. To address the challenges posed by concept drift and the high cost of retraining, Attarian *et al.* [64] introduced AdaWFPA, an adaptive online WF attack that overcomes the limitations of static models, whose accuracy degrades over time as website traffic patterns change. AdaWFPA leverages adaptive stream mining algorithms, Adaptive Hoeffding Tree (AHT), and Adaptive Hoeffding Option Tree (AHOT), to dynamically update its classifier in real-time, ensuring robust performance against evolving website characteristics. The method is evaluated on Tor traffic datasets, demonstrating superior accuracy, precision, and recall compared to state-of-the-art static attacks, even when tested against defenses like Walkie_Talkie [45].

2) DL-based WF Attack models: Traditional ML models are simpler and more interpretable, making them easier to analyze and deploy with fewer resources. But DL models outperform traditional ML models in terms of accuracy and robustness, particularly in open-world scenarios and in the presence of defenses. These DL-based attacks leverage deep neural networks’ powerful feature extraction capabilities to automatically learn complex patterns in encrypted network traffic, enabling more accurate identification of the websites a user is visiting through the Tor browser. TABLE VII summarizes representative deep learning based WF models. In this paper, the DL-based WF attack models will be mainly classified into 4 categories: AutoEncoder (AE) based WF attacks, Convolutional Neural Networks (CNNs) based WF attacks, Recurrent Neural Networks (RNNs) based attacks, and Few-shot learning based WF attacks. To highlight the increasing use of Few-shot learning techniques, we provide a separate table summarizing the existing works.

AutoEncoder-based WF Attacks: In 2016, Abe *et al.* [10] first introduced DL into WF attacks, and proposed a fingerprinting attack based on the Stacked Denoising Autoencoder (SDAE) mechanism. The attack identifies websites accessed by users through analyzing traffic direction, even when the traffic is encrypted. This paper demonstrates the great potential of DL techniques in increasing the WF accuracy and achieves

TABLE VII: A summary of representative deep learning based WF models.

Research	Architecture	Dataset	Closed World		Open World		
			Dataset Size	Acc (%)	Dataset Size	TPR (%)	FPR(%)
SADE [10]	SADE	Wang14 [9]	100×90	88.00	9,000×1	86.00	2.00
	SDAE			94.25		71.30	3.40
AWF [94]	CNN	AWF [94]	900×2,500	91.79	500,000×1	70.94	3.82
	LSTM			88.04		53.39	3.67
p-FP [112]	CNN	Wang14 [9]	NA	NA	90×100	96.72	1.78
	MLP			NA		96.56	2.78
DF [93]	CNN	DF [93]	95×1000	98.30	40,716×1	95.70	0.70
Var-CNN [96]	CNN	AWF [94]	900×2,500	97.80	500,000×1	89.20	1.10
Tik-tok [93]	CNN	DF [93]	95×1000	98.40	40,716×1	94.00	NA
He-GRU [152]	ResNet and GRU	AWF [94]	900×2,500	99.10	500,000×1	96.90	0.71
2ch-TCN [122]	CNN	Wang14 [9]	100×90	93.73	NA	NA	NA
WF-Transformer [70]	Transformer	DF [93]	95×1000	99.10	40,716×1	96.90	0.70
snWF [125]	Neural network ensemble	AWF [94]	NA	NA	400,000×1	98.10	5.70
RF [119]	TAM and CNN	DF [93]	95,000	96.77	40,000	>85.00	NA
TinyWFP [119]	Wavelet transform and Lightweight Neural Network	AWF [94]	2,500	99.20	20,000	97.50	4.50
		DF [93]	1,000	98.10			
WFCAT [116]	IAT histogram and CNN	WFCAT [116]	100×100	94.47	NA	NA	NA
CF [140]	Contrastive learning and Few-shot	Wang14 [9]	100×90	90.40±10.00	DF-9,000	91.20	7.40
		AWF [94]	95×100	88.40±10.00			
		DF [93]	100×90	87.10±10.00			

an 88% accuracy with the closed-world dataset. Later, Rimmer *et al.* [94] replaced the manual, expert-driven process of selecting traffic features with automatic feature learning via SDAE, CNN, and LSTM. The automated website fingerprinting (AWF) enables the attack to adapt better to traffic obfuscation techniques and changes over time. This approach also increases the practicality and stealth of WF attacks, especially in scenarios involving dynamic content or large-scale open-world environments. At the same time, Se Eun *et al.* [112] demonstrated that unsupervised DNNs, specifically autoencoders, can generate low-dimensional feature representations that improve the performance of the SOTA WF attacks. These AE-generated features reduce computational costs while maintaining or enhancing classifier accuracy.

CNN-based WF Attacks: CNNs are originally designed for computer vision tasks, and have emerged as a powerful tool in

WF attacks. Their ability to automatically extract hierarchical features from raw input makes them particularly effective at learning subtle patterns in encrypted or anonymized network traffic. In 2018, Sirinam *et al.* [93] introduced Deep Fingerprinting (DF), a CNN-based architecture designed to classify Tor traffic without requiring handcrafted features. DF achieves over 98% accuracy on their collected undefended Tor traffic, outperforming all prior WF attacks. In the same year, Bhat *et al.* [96] proposed Var-CNN, which combined CNNs with automated cumulative feature extraction (*e.g.*, total packet counts, transmission time). Leveraging ResNet-18 and dilated causal convolutions, Var-CNN captured long-term dependencies in packet sequences, showing that combining packet direction and timing improves classification accuracy.

Rahman *et al.* [12] later introduced burst-level timing features, which capture temporal relationships within and between

bursts of traffic. This led to Tik-Tok, a WF attack using directional timing representations that combine packet timing and direction. To further exploit timing, Wang *et al.* [122] proposed a two-channel Temporal Convolutional Network (2ch-TCN) that separately models packet direction and timing sequences. On the Wang14 dataset [9], 2ch-TCN achieved 93.73% accuracy, outperforming CUMUL [11], Rimmer-CNN/SDAE [94], and Var-CNN [96]. Building on this, Shen *et al.* [153] developed BurNet, a fine-grained CNN-based WF method that processes unidirectional burst sequences rather than bidirectional packet sequences. BurNet achieved a precision and recall of 99.00%, making it effective in both local and remote attack settings. While defenses such as packet padding, traffic regularization, and delaying techniques reduce WF attack accuracy, Shen *et al.* [119] proposed Robust Fingerprinting (RF), which maintains strong performance under defenses. RF introduces a Traffic Aggregation Matrix (TAM) representation that encodes both packet direction and timing, allowing a CNN classifier to learn defense-resilient fingerprints. RF achieved an average accuracy improvement of 8.9% over Tik-Tok [93] against existing defenses.

To address the computational cost of DL-based WF attacks, Tian *et al.* [131] introduced Tiny WFP, a lightweight method that applies wavelet-based dimensionality reduction and depthwise separable CNNs. By retaining only low-frequency wavelet coefficients, Tiny WFP achieved 99.2% accuracy in closed-world settings while being $81\times$ smaller and $79\times$ more efficient than DF. In parallel, Pan *et al.* [123] tackled the problem of limited training data. They combined CNNs with transfer learning to stabilize performance when traffic statistics vary irregularly over time. Their model integrated packet direction with timing sequences, significantly improving accuracy in few-shot scenarios. Gong *et al.* [116] argued that existing attacks underutilize timing features, which remain a persistent leakage source. They proposed WFCAT, which introduces an Inter-Arrival Time (IAT) histogram representation and CNN architecture with novel timing-aware blocks. WFCAT achieved over 59% accuracy against Surakav, outperforming RF and Tik-Tok by 28% and 48%, respectively, in the closed-world setting.

Despite DL-based WF attacks achieving high accuracy, existing attacks suffer from high false positive rates, high computational cost, and low sensitivity. Chawla *et al.* [41] proposed ESPRESSO to improve the accuracy and efficiency of flow correlation attacks while reducing false positive rates and computational complexity compared to existing approaches. ESPRESSO utilizes an aggregated feature representation and employs Transformers for global processing to capture long-range dependencies. This method significantly outperforms DeepCoFFEA [154], especially in terms of reducing false positives and improving correlation accuracy.

More recently, Zou *et al.* [40] introduced Relation-CNN, which enhances ResNet18 through three optimizations to preserve data fidelity and a feature-attention block to weight channel contributions. Relation-CNN improved prediction accuracy by up to 5% and 8% on defended and undefended datasets, respectively. To address the significant reduction in classification accuracy from trace adversarial examples,

Shi [39] introduced a multiscale fingerprinting (MF). To effectively recognize the complex real patterns of anonymous traces under defenses, MF enhances feature extraction in two key ways. First, it slices each trace using multiscale time slots to construct multiscale trace matrices as inputs, thereby enriching temporal information. Second, it utilizes a sophisticated CNN structure with multiscale convolution layers to extract complicated features from changed patterns of varying lengths. Deng *et al.* [32] introduced Holmes, an early-stage WF attack that accurately identifies visited websites even when only a small portion of the page has loaded. Traditional DL-based WF attacks require complete page-loading traffic, making them unreliable under dynamic network conditions and WF defenses. Holmes overcomes this limitation by leveraging spatial-temporal distribution analysis to correlate early-stage traffic with pre-collected full traffic, significantly improving attack robustness.

To address the concept drift problem, Xu *et al.* [42] introduced APWF, a parallel WF attack method that integrates an attention mechanism to address challenges like concept drift and small-sample inefficiency in traditional WF attacks. The APWF model combines CNN and BiGRU to capture spatial and temporal features of encrypted traffic, enhanced by a channel attention mechanism (CBAM) for improved feature learning. Additionally, the authors propose a fine-tuning method based on transfer learning to adapt to evolving website patterns with minimal labeled data. The method can maintain the accuracy at 92.4% in the scenario of 56 days between the training data and the target data, which is only 4% less loss compared to the instant attack.

RNN-based WF Attacks: RNNs are designed to capture temporal dependencies in sequential data, making them suitable for modeling the time-series nature of network traffic, such as packet sequences in website visits. Compared to the outstanding performance of Autoencoders (AE) and CNN in the field of WF attacks, RNNs usually perform poorly and have higher time complexity [38], [94]. However, RNNs are effective at capturing the temporal characteristics of website fingerprints and are often used in combination with CNNs. He *et al.* [152] introduced a complex deep neural network to automatically learn features and classify website fingerprints. The model uses a two-layer GRU network to extract the time feature and a 50-layer residual network to extract the spatial features of the website fingerprint. This method can achieve a classification accuracy rate of over 99%.

Few-Shot learning based WF Attacks: Few-shot learning is a DL approach where a model learns to recognize new classes using only a few labeled examples per class. It is widely used in few-shot classification and meta-learning, especially when collecting large labeled datasets is expensive or impractical, and the environment (like websites or services) changes frequently. In WF attacks, traditional DL models (*e.g.*, CNN-based WF attacks) require many traffic traces per website/webpage to train accurately. However, websites change layout/content frequently, and collecting large amounts of new traffic for retraining is time-consuming. Moreover, attackers might only get access to a few traces per target page. TABLE VIII demonstrates the existing representative

TABLE VIII: A summary of representative Few-shot learning based WF models.

Research	Dataset	Closed World (Acc %)					Open World (Pre %)			
		1-shot	5-shot	10-shot	15-shot	20-shot	5-shot	10-shot	15-shot	20-shot
TF [95]	AWF [94]	79.20 $\pm$ 1.30	92.20 $\pm$ 0.60	93.90 $\pm$ 0.20	94.40 $\pm$ 0.30	94.50 $\pm$ 0.20	87.10	90.80	89.10	87.30
TForm-RF [130]	AWF [94]	NA	86.30 $\pm$ 1.00	93.60 $\pm$ 0.20	95.70 $\pm$ 0.30	96.50 $\pm$ 0.30	92.20	95.30	96.80	97.70
JAN [136]	AWF [94]	95.60 $\pm$ 0.10	98.30 $\pm$ 0.00	98.70 $\pm$ 0.00	98.80 $\pm$ 0.00	98.90 $\pm$ 0.00	97.20	98.10	98.10	98.50
TrafficSiam [137]	AWF [94]	NA	85.22	89.52	91.31	92.32	99.39	98.44	98.17	96.99
WF3A [139]	Tik-Tok [12]	68.89	85.33	87.53	88.28	88.86	86.63	90.12	89.64	89.24
SMC [138]	Mixed ([9], [57], [94])	59.11	76.20	80.70	82.37	82.61	78.96	83.69	NA	NA

few-shot learning based WF attacks. The research works and evaluation results are selected from datasets that contain both closed-world and open-world settings.

Sirinam *et al.* [95] first explored WF in Tor by proposing Triplet Fingerprinting (TF), a WF attack that leverages N-shot learning (NSL) to significantly reduce the amount of training data required. NSL allows the attacker to identify websites using only a few examples (*e.g.*, 5 examples per website), making the attack more practical and portable. The base model refers to the CNN model that is used as the sub-network in the triplet networks. The K-NN model is implemented directly to measure the similarity between two different samples. This approach reduces the need for large datasets and frequent retraining, and achieves approximately 90% precision and 80% recall when tuned for precision. However, there is a large set of relevant auxiliary training samples for the model, which makes it unrealistic. To overcome this limitation, Chen *et al.* [97] improved Sirinam’s work by introducing a model-agnostic, efficient, and harmonious data augmentation (HDA) method, which aims to identify the websites a user visits by analyzing encrypted network traffic patterns, even in privacy-preserving networks like Tor. The HDA method offers three different data augmentation operations, including random rotation, masking, and mixing in intra-sample and inter-sample fashion. This improved attack model increases the accuracy from 87% to 90.60% under the dataset of Wang14 [9]. However, this method is unsatisfactory in small training data cases (*e.g.*, less than 10 training samples per class). To overcome this limitation, Transfer Learning Fingerprinting Attack (TLFA) [134] was proposed by training a strong embedding model and leveraging classical machine learning models (*e.g.*, SVM) on top. But this method remains inferior in the 1-shot case, especially for the open-world setting. Then, Chen *et al.* [128] introduce the understudied problem of few-shot WF attacks, emphasizing the need for adaptability to new websites with minimal labeled data. The paper addresses the challenge of few-shot WF attacks, where the goal is to identify visited websites from encrypted traffic data using only a few labeled training samples per website. This is a more realistic and scalable scenario compared to traditional WF attacks that assume abundant training data.

The new few-shot model enhanced the accuracy up to 93% compared to previous works.

Collecting large-scale and fresh traces is quite costly and unrealistic for DL-based WF attacks. When new traffic traces with different distributions or newly added monitored websites are introduced, DL-based WF attacks require a long bootstrap time for retraining. To address this limitation, Zou *et al.* [132] introduced a cross-domain, low-data WF attack called WFBDC (WF with Brownian Distance Covariance). WFBDC advances WF attacks by enabling efficient cross-domain few-shot learning with historical data. Its BDC-based approach enhances accuracy and reduces training time, demonstrating practicality for real-world deployment. Chen *et al.* [130] proposed a data augmentation technique that generates additional virtual traffic samples by extracting Trace-Forms (TForms), patterns derived from the structure of cell sequences in Tor traffic. By randomly refilling these TForms, they create new and meaningful samples that expand training data distribution. Through extensive experiments in both closed-world and open-world settings, TForm-RF is shown to significantly improve the classification accuracy of deep-learning-based WF attacks. Then, Xie *et al.* [140] introduced Contrastive Fingerprinting (CF), a WF attack leveraging contrastive learning and data augmentation to achieve high accuracy with minimal training traces. CF utilizes contrastive learning to produce a feature extractor in the pre-training stage. Then, the few-shot stage feeds the feature extractor with augmented traces and trains a classifier for testing users’ traces. Their work achieves a high accuracy of 90.4% in the closed-world scenario and distinguishes monitored websites with a high True Positive Rate of 91.2% in the open-world scenario. The previous few-shot-based WF attacks adopt complex metric strategies or perform time-consuming transfer learning, neither of which yields the most efficient performance in dynamic network environments. To address this issue, Lyu *et al.* [135] introduced Task-adaptive Feature Alignment Network (TFAN), a meta-learning-based approach for few-shot learning on WF attacks. TFAN regards the few-shot learning approach as a feature alignment problem in the class latent space, aiming to depict each location in the query feature map as a weighted sum of support features of a given class.

Although existing few-shot WF attacks have shown some progress in addressing the data scarcity problem in WF attacks, they still have some limitations. Triplet Fingerprinting (TF) has a complex training process and does not fully leverage large auxiliary training data; TF and CWFA only focus on the feature space alignment between the originally and newly collected traces. TLFA and MBL methods use a fine-tuning process to obtain a task-specific classifier; however, they ignore the feature space alignment for the originally and newly collected traces. Zhou *et al.* [136] introduced a new few-shot WF attack method called Joint Alignment Networks (JAN) to address the limitations of existing methods in fine-grained feature alignment to improve the performance in few-shot WF attacks by jointly aligning the feature space and semantic space for originally and newly collected traces. This method enhances few-shot website fingerprinting (FS-WF) attacks, where an adversary aims to classify web traffic traces with minimal labeled data. Extensive experiments on public datasets show that JAN outperforms the state-of-the-art few-shot WF methods, especially in the difficult 1-shot tasks. Wang *et al.* [137] introduced TrafficSiam, a few-shot learning based WF attack that leverages contrastive learning and self-supervised pre-training to classify encrypted Tor traffic with minimal labeled data. Unlike traditional DL-based attacks [93]–[96], [119] that require extensive labeled datasets, TrafficSiam pre-trains on unlabeled Tor traffic to learn robust feature representations, then fine-tunes with just a few labeled samples. This method achieves 92.32% accuracy in the closed-world setting. Wang *et al.* [139] introduce WF3A, a N-shot learning based WF attack that enhances attack accuracy in few-shot learning scenarios. Website fingerprinting is a privacy risk, allowing adversaries to identify websites visited by users over encrypted networks. Different from previous few-shot works, this work innovatively combines direction and length, designs and uses the fusion feature that covers more website identification information. To interpret the traffic patterns that can identify the website categories from the fusion feature, they introduce the Enhanced Channel Attention (ECA) and design a stronger feature extractor when constructing the WF model, which enhances the learning ability of the model by the channel attention mechanism.

Traditional Few-Shot Website Fingerprinting (FSWF) attacks struggle with adaptability due to limited training data diversity [95], [132], [134], [139], leading to poor generalization across different network conditions. Tan *et al.* [138] introduced a joint learning framework to achieve the collusion FSWF attacks. The proposed method fuses the feature spaces of multiple user-side attackers to enhance the representation ability of the local model, and constructs a virtual fusion center to mitigate the impact of Non-IID (Non-Independent and Identically Distributed) data. Compared with the state-of-the-art method, the proposed method improves the accuracy by up to 13.02% in the closed-world setting and the AUC by up to 8.50% in the open-world setting, respectively. To address the challenge of WF in cross-environmental settings, where differences in browsers and proxy software cause feature drift and hinder model performance. Li *et al.* [141] proposed X-EPRINT, a systematic framework designed to generate cross-

environmentally invariant features, apply potential-aware traffic resampling, and use inter-flow data augmentation for few-shot adaptation. The method tackles three core challenges: feature drift, the sampling dilemma, and few-shot generalization. Experimental evaluation across 18 scenarios shows that X-EPRINT achieves an F1-score of 0.719 in zero-shot recognition, 58.4% higher than the best baseline, and 0.925 in 3-shot adaptation, demonstrating strong robustness. However, its performance diminishes under severe feature drift, particularly with Firefox traffic, and the system still requires some resampling in new environments for optimal accuracy.

Hybrid or other deep learning based WF Attacks: To enhance low-data training, Oh *et al.* [143] introduced Generative Adversarial Networks (GANs) for Data-Limited Fingerprinting (GANDaLF), which leverages GANs to generate synthetic data, enabling effective training of a deep neural network even with few real training samples. The paper evaluates GANDaLF in low-data scenarios, including fingerprinting both website index pages and non-index subpages, and achieves closed-world accuracy of 87% with just 20 instances per site (and 100 sites) in standard WF settings. In particular, GANDaLF outperforms TF by a 29% margin and Var-CNN by 38% for training sets using 20 instances per site. Guo *et al.* [87] introduced a homology analysis-based WF attack model, relying on a Convolutional Neural Network-Bidirectional Long Short-Term Memory (CNN-BiLSTM), targeting SSH and Shadowsocks encrypted traffic in anonymized networks. The method innovatively employs one-shot matrices to represent traffic features, enhancing fault tolerance and feature dimensionality, and achieves 94.8% and 98.1% accuracy in classifying SSH and Shadowsocks anonymous encrypted traffic.

In 2022, Chen *et al.* [129] found that the current deep learning model is data-hungry when simulating the mapping relations of traffic and the website it belongs to, which may not be practical in reality. Then, they introduced the Send-and-Receive Pair (SRP) as a fundamental unit to analyze and generate bionic traffic traces for enhancing WF attacks on Tor. Their approach improves the Var-CNN model, achieving up to 50% higher accuracy in few-shot settings and outperforming transfer learning methods like TF in closed-world scenarios (95% accuracy). Wang *et al.* [125] analyzed the existing deep learning-based WF attack models and found that these attacks are sensitive to the specific information in the data and would result in less-than-ideal performance on data outside the training set or on data that is impacted by the effect of concept drift. They introduced snWF, which leverages an ensemble of deep learning models to improve the robustness and accuracy of identifying websites visited by users over encrypted connections. snWF introduces an ensemble technique called snapshot ensemble, where multiple snapshots of a deep neural network are taken during a single training run. These snapshots are then combined to reduce the variance of the model, making it more robust to concept drift and data outside the training set. This method achieves a TPR of 98.10% and a FPR of 5.70% with 400,000 open-word websites.

Singh *et al.* [142] presented AutoKeras, a neural architecture search (NAS) tool, to explore whether automated DNN architecture design can improve WF attacks compared

to manually crafted models. While NAS-generated models require larger exploration budgets and longer inference times, they highlight the potential for automated deep-learning-driven WF attacks. Zhuo *et al.* [70] introduced WF-Transformer, the first WF attack model based on Transformer networks, which is designed to capture the long-term dependencies in traffic traces, a feature that traditional CNN-based models fail to extract. The model uses packet direction as input and employs a sophisticated design of the Transformer network, including relative position embedding and multi-head attention mechanisms, to improve feature extraction. Jansen *et al.* [155] investigated the real-world feasibility of WF attacks on Tor by addressing a critical gap in previous research: the mismatch between training data collected from Tor exit relays and real-world WF attacks conducted from entry relays. To bridge this gap, the authors introduce Retracer, which transforms exit-side traces into entry-side traces, improving classifier training and yielding more accurate real-world WF threat estimates.

To address the limitation associated with WF attacks and defense strategies in the presence of noise, Huang *et al.* [156] proposed a filter-assisted attack (FAA) to counter random packet defenses (RPD) and introduced a list-assisted defense (LAD) to mitigate FAA, balancing defense efficacy and overhead. The next year, Qian *et al.* [124] investigated the impact of noise on WF attacks and introduced two attack methods: filter-assisted attack and augmentation-assisted attack. The first attack method leverages the packet size distribution to effectively filter out noise, while the second one trains a classification model by incorporating artificial noise. Experimental results show the efficacy of defending against different attacks and noises. Bahramali *et al.* [71] identified that WF classifiers often fail when deployed in unobserved network conditions due to variations in bandwidth, Tor circuits, and traffic patterns. To overcome this limitation, they proposed a network trace augmentation technique, NetAugment, to enhance the robustness of WF models. The NetAugment was instantiated through semi-supervised and self-supervised learning techniques.

Through analyzing the present WF attacks, Mathews *et al.* [30] found that a multi-channel input format that provides richer contextual information and enables the model to learn robust representations even in the presence of heavy traffic obfuscation. Then, the authors introduced Laserbeak, a state-of-the-art WF attack designed to break Tor defenses by leveraging multi-channel feature representation and transformer-based deep learning models. LASERBEAK demonstrates absolute performance improvements of up to 36.2% compared with prior attacks against defended traffic.

The deep learning based WF attacks often ignore low-level timing information and use a simple direction-based representation method that only retains the spatial information of website traffic packets. However, the timing information plays a significant role in the WF attack under the research of Hong *et al.* [38]. They presented a WF attack technique called Time-Sampling, which leverages both spatial and temporal timing information from website traffic to improve the accuracy of WF attacks. The Time-Sampling model uses a time-based sampling technique that fills packet gaps with zeros, preserving

both timing and spatial information in the traffic. Through evaluation, the accuracy is improved by 20% at most, higher than direction-based attacks.

Traditional WF techniques either rely on single-flow features, which are sensitive to network changes, or multi-flow aggregation methods that fail to capture temporal and spatial relationships between network flows. To solve this issue, Tan *et al.* [75] presented STC-WF, a WF attack that leverages spatio-temporal correlations among network flows to improve classification accuracy, particularly in encrypted traffic scenarios. The STC-WF uses a Graph Neural Network (GNN)-based approach that models these correlations as a Spatio-Temporal Correlation Graph (STCG). To overcome the limitation of traditional WF techniques, which struggle with content updates and labeling overhead, limiting their effectiveness in real-world scenarios. Gao *et al.* [31] introduced MRCGCN, a self-supervised learning-based WF attack that leverages multi-level resource-coherent GNNs to improve the accuracy and robustness of WF attacks. MRCGCN overcomes these challenges by representing website traffic as structured graphs at resource, flow, and host levels and applying graph learning techniques for classification.

Concept drift in website fingerprinting can occur when websites modify their structure or content. Even small changes, such as adding images, updating ads, or reorganizing page elements, alter the resulting traffic patterns. These shifts cause the distribution of features like packet sizes, timing, and traffic length to deviate from the data on which models were trained. To address the problem of concept drift, Ding *et al.* [44] presented LRCT, a DL-based WF attack model designed to improve classification accuracy and efficiency while addressing concept drift—a phenomenon where model performance degrades over time due to network changes. The LRCT network effectively leverages the temporal learning advantages of Local RNN and the spatial learning strengths of CNN by designing the local feature extraction block (denoted as LRC Block). The network then uses a Transformer Encoder to capture more robust global features from the low-dimensional data. Experimental results comparing LRCT with state-of-the-art WF methods demonstrate that the proposed LRCT achieves 99.34% accuracy in a closed-world scenario, outperforming other benchmark models.

Ginige *et al.* [157] proposed TrafficGPT, the first solution that leverages GPT-2 for feature extraction in encrypted traffic classification. GPT-2's ability to understand sequential data is utilized to extract comprehensive features, improving the performance of open-set classification. This paper focuses on the open-set scenario, where the model must distinguish between known and unknown classes. This is particularly important in real-world applications, such as detecting harmful content while allowing benign traffic. Yuan *et al.* [148] introduced Class Incremental Fingerprinting (CIF), a WF attack designed to continuously learn and expand its classification capabilities in real-world dynamic environments. Traditional DL-based WF attacks achieve high accuracy in static experimental settings but struggle in open-world scenarios, where new websites constantly emerge. CIF addresses this limitation by incorporating Class Incremental Learning (CIL) techniques, allowing

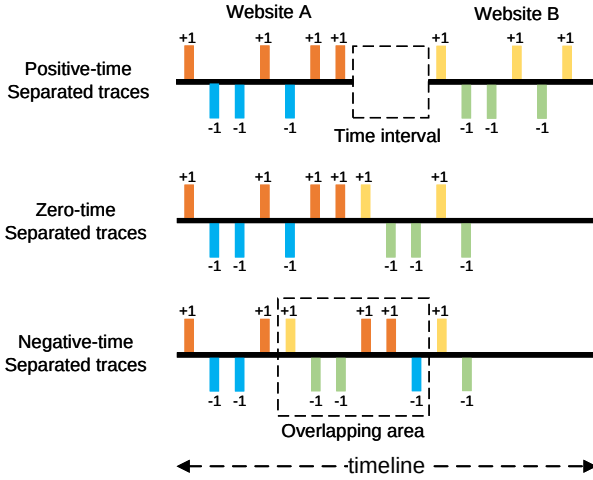


Fig. 6: The three types of basic multi-tab traces in the 2-tab setting. When a user accesses website B in a new tab before website A has fully finished loading, it creates overlapping segments in the resulting multi-tab traces generated by the browser.

the attack model to adapt to new monitored websites without forgetting previously learned knowledge. Yang *et al.* [147] introduced DTPN (Diffusion-based Traffic Purification Network), which counteract adversarial perturbations in Tor WF defenses by leveraging diffusion models. Unlike traditional adversarial training, which struggles with generalization and frequent updates, DTPN purifies protected network traffic by removing adversarial noise, enabling existing WF classifiers to maintain high accuracy without retraining. The proposed UNetFormer architecture, combining TransformerBlocks and one-dimensional convolutions, estimates noise effectively, while a custom loss function enhances model generalization. Mei *et al.* [146] presented HPETC, a High Precision and Efficient Traffic Classification system designed to classify Tor traffic in real-world, large-scale network environments. Unlike previous Traffic Classification (TC) and WF techniques, HPETC effectively mitigates the base rate fallacy, a common issue where false positives overwhelm the classification process due to the extremely low proportion of anonymous traffic compared to normal network traffic.

3) *Multi-tab WF attacks*: Traditional WF attacks are performed on a well-constructed dataset, in which each packet trace is generated by visiting only one page tab at a time. Thus, attackers only need to analyze traces entirely belonging to a specific website without any noise data, which is known as the single-tab assumption. This assumption makes WF attacks less difficult, but it is unrealistic: users may engage in other activities, such as instant messaging alongside web browsing, and similar behavior is entirely random [37]. In addition, even if users only browse the web, they may open a new page tab before the last one finishes loading, a behavior known as multi-tab browsing [6], [80], [158], such as jumping to another page through a hyperlink. A comparison of representative multi-tab WF attacks with 10% overlapping proportions as shown in TABLE IX.

After Juarez *et al.* [6] pointed out that multi-tab browsing behavior significantly degrades the performance of WF models under the single-tab assumption, Wang and Goldberg [37] attempted to transform multi-tab traces into single-tab traces

by splitting packet sequences and processing them using existing single-tab WF classifiers. In [37], multi-tab traffic traces are categorized into three fundamental types based on their time interval relationships: Positive-time, Zero-time, and Negative-time. Fig. 6 illustrates these scenarios for 2-tab traffic traces: (1) **Positive-time separated traces** refer to instances where the user lingers on a web page for a period before navigating to the next one, resulting in a brief lull in network activity and a noticeable gap in traffic. This time gap is significant enough to be detected as a pause but still shorter than the threshold used in time-based traffic splitting. (2) **Zero-time separated traces** refer to scenarios where a user clicks a link on a web page that is still loading, interrupting the current page and immediately initiating requests for the next one. This results in a clear transition between two web pages without any noticeable time gap between them. (3) **Negative-time separated traces** refer to situations where the user is loading two web pages simultaneously in multiple tabs. In such cases, the appropriate split point is considered to be the moment the second page begins loading. This is the most challenging class to separate, as there is neither a noticeable time gap nor a distinct cell pattern signaling the transition. Nonetheless, machine learning techniques can still be used to identify the split by extracting informative features from the cell sequences.

Wang and Goldberg [37] focused on identifying split points within multi-tab traffic to divide it into multiple different single-tab segments for classification. However, this segmentation process results in information loss in the overlapping portions of the traffic. Although the algorithm improves the accuracy of split-point detection, it can only predict the first webpage visited in a multi-tab browsing session. Moreover, as the number of tabs increases, accurately locating the split points becomes increasingly difficult [120]. Xu *et al.* [62] introduced the BalanceCascade-XGBoost algorithm to identify the split point between two tabs and classify traffic according to the initial chunk of data. Later, Yin *et al.* [65] improved the BalanceCascade-XGBoost algorithm by applying Split Point Identification and Chunk-Based Classification mechanism. Cui *et al.* [80] utilized the Hidden Markov Model to segment continuous browsing traffic and conducted webpage recognition on each segment through K-NN, achieving an impressive 80% accuracy in identifying the split points. To determine the final category of the webpage, a majority vote was employed. However, it treats overlapping areas equally with other areas and fails to consider the relevance and importance of the overlapping area in determining the webpage category [144].

In 2020, Cui *et al.* [121] utilized a CNN model to differentiate between single-tab and multi-tab traces, and extracted a fixed number of packets from the beginning and end of 2-tab traces. They then used CNN, Long Short-Term Memory (LSTM), and Stacked Denoising Auto-Encoding (SDAE) models to classify the extracted pure trace segments, further improving the accuracy of WF. However, their approach is still constrained by a limited number of page configurations and relies on multi-tab traces containing sufficiently long and non-overlapping pure segments. Then, Guan *et al.* [118] proposed a block attention profiling model (BAPM), which

TABLE IX: A summary of representative multi-tab WF attacks with 10% overlapping proportion.

Research	Architecture	Dataset Name	Closed World Precision(%)				Open World Precision(%)			
			2-tab	3-tab	4-tab	5-tab	2-tab	3-tab	4-tab	5-tab
BAPM [118]	Self-attention and CNN	Xu et al. [62] and Manually Merged BAPM [118],	66.80	77.30	NA	NA	84.90	NA	NA	NA
TMWF [43]	Transformer and CNN	Walkie-Talkie [45] and Manually Merged	95.50	NA	NA	NA	>70.00	~70.00	>70.00	>70.00
ARES [110]	Top-k multi-head self-attention and CNN	ARES [110]	91.30	91.30	90.20	81.80	75.20	58.70	52.80	43.80
FingerMamba [127]	Mamba model and CNN	ARES [110] and Manually Merged DS-19 [57],	83.00	80.80	79.70	75.70	81.00	79.00	>75.00	>75.00
MTWFAs [13]	FCN	ARES [110] and Manually Merged	68.60	68.60	70.00	72.10	73.30	74.80	73.90	71.80
FMWF [133]	Few-shotFine-tuning	ARES [110] and Manually Merged	88.50	77.90	89.00	84.90	NA	NA	NA	NA
MW3F [120]	Transformer and CNN	ARES [110], Deng et al. [32], and Manually Merged	92.91	93.11	92.66	89.88	>90.00	>90.00	>90.00	>90.00
CountMamba [29]	State Space Model and CNN	ARES [110] and TMWF [43]	87.33	81.52	81.26	73.89	85.02	81.17	79.98	75.60

focuses on overlapping webpage traffic. However, with the number of webpages increasing, BAPM faces the challenge of dynamically adapting the number of attention heads, which introduces instability to the network structure and leads to more parameters. In addition, BAPM only considers a single scenario of overlapping visits and cannot adapt to more complex browsing scenarios. Previous CNN-based WF attacks rely on a carefully curated dataset where each packet trace is created by loading only one webpage at a time. Under this single-tab assumption, attackers analyze clean, noise-free traces, each belonging exclusively to a single website. However, the assumptions of previous work are obviously unreal. Users may access multiple webpages in multiple tabs at the same time.

Guan *et al.* [118] found that the existing multi-tab attacks fail to exploit the overlapping regions, which contain mixed data from adjacent page tabs. To address this limitation, they leveraged the CNN to generate a tab-aware representation on the whole packet trace, including overlapping areas, and split the representation into blocks to reduce the influence of mixed data under the traffic direction features. Using multi-head attention, the model groups blocks belonging to the same website, allowing it to identify multiple websites simultaneously from a global perspective. This approach eliminates the need for explicit feature engineering and adapts naturally to different degrees of overlap.

Later, Jin *et al.* [43] introduced Transformer-based Multi-tab Website Fingerprinting (TMWF), an end-to-end deep learning

model that leverages a Transformer architecture to improve multi-tab WF attacks. Unlike previous models, TMWF does not require prior knowledge of the number of tabs in a browsing session. It treats multi-tab WF recognition as a set prediction problem, similar to object detection models. By introducing queries for tabs to extract WF features of different tabs and new evaluation metrics, TMWF achieved high-accuracy identification with fewer than six tabs. Deng *et al.* [110] introduced ARES, which is designed to perform WF attacks in multi-tab browsing scenarios. ARES formulates the multi-tab WF attack as a multi-label classification problem, where each classifier identifies whether a specific website is visited based on local traffic patterns. It uses a transformer-based model (Trans-WF) to extract and analyze local traffic patterns from multiple short traffic segments, even when the overall traffic pattern is obfuscated by WF defenses. In addition, ARES does not require prior knowledge of the number of open tabs, making it more practical for real-world deployment. Xie *et al.* [144] proposed a multi-scene WF method based on multi-head attention and data enhancement to address challenges in multi-tab and incomplete webpage traffic scenarios. For multi-tab browsing, the method (MAMWF) employs sequence embedding and block division to preserve traffic order, followed by multi-head attention to extract global features from mixed regions, improving classification accuracy. For incomplete traffic, the data enhancement-based webpage fingerprinting (DEWF) method uses generative adversarial imputation nets (GAIN) to enhance missing data, restoring

fingerprint information for better recognition. Cui *et al.* [145] introduced Enhanced Deep Fingerprinting (EDF), a pair of DL-based WF attacks to overcome zero-delay lightweight defenses like FRONT and GLUE. The authors propose two models: EDF-S which integrates a global average pooling layer and a series structure into the existing Deep Fingerprinting (DF) framework, and EDF-P, which employs a parallel model for enhanced feature extraction.

In 2024, Liu *et al.* [127] introduced FingerMamba, an approach to multi-tab WF using the Mamba-based structured state-space model (SSM). FingerMamba addresses the more realistic multi-tab scenario, where users open multiple tabs, increasing classification complexity. Yin *et al.* [13] investigated multi-tab website fingerprinting attacks (MTWFAs) using deep learning to analyze encrypted Tor traffic, focusing on realistic scenarios where users browse multiple websites simultaneously. The paper introduces MTWFA-SEG, an attack method that classifies each packet in multi-tab Tor traffic, enabling detailed inference of browsing activities and timing. The authors propose an improved fully convolutional network (FCN) model with skip transformations, residual connections, and a smoothing regularization technique to handle varying-length traffic traces effectively. Zhao *et al.* [33] presented Oscar, a WF attack framework designed to identify fine-grained webpages (*e.g.*, subpages of the same website) rather than just websites, a challenge that traditional WF attacks struggle with. Existing WF attacks primarily classify entire websites, but their effectiveness drops significantly when trying to distinguish between different webpages due to high traffic similarity and large-scale dataset challenges. Unlike previous WF methods that focus on websites, Oscar applies metric learning to differentiate between similar-looking webpages by transforming the feature space. Chen *et al.* [159] presented RobustWF, a WF attack designed to handle multi-tab web browsing over encrypted tunnels. Traditional WF techniques struggle in realistic network environments due to dynamic tab concurrency, packet loss, duplication, and disorder. The proposed RobustWF method improves website fingerprinting accuracy and robustness using causality correlation and context learning. Instead of treating traffic as a mixed sequence, RobustWF groups packets into causality chains based on user requests and server responses. These chains help separate traffic from different websites, mitigating the tab concurrency challenge.

In 2025, Meng *et al.* [133] presented Few-shot Multi-tab Website Fingerprinting (FMWF), an approach to multi-tab WF attacks that significantly reduces the need for large training datasets and frequent retraining. Unlike traditional WF models that require vast amounts of real-world multi-tab data, FMWF leverages transfer learning to adapt pre-trained models to multi-tab environments using only a small set of real traffic samples. Yuan *et al.* [120] introduced MW3F, a multi-tab WF attack that leverages Transformer-based feature fusion to enhance attack accuracy and robustness in encrypted networks like Tor. Traditional single-tab WF attacks are ineffective against multi-tab browsing, where traffic from different websites overlaps, making website identification difficult. MW3F overcomes this by integrating multi-head

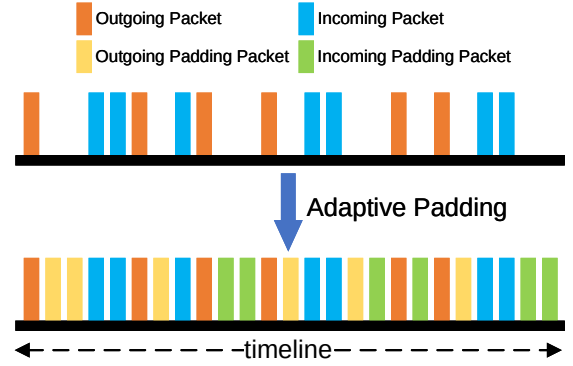


Fig. 7: An overview of the adaptive padding mechanism, which inserts dummy cells into both incoming and outgoing traffic and introduces delays to packets on either the user side or the bridge side to obscure traffic patterns.

self-attention and learnable label embeddings to adaptively extract discriminative traffic features, even in the presence of strong WF defenses. Deng *et al.* [29] introduced CountMamba, a generalized WF attack framework that effectively identifies Tor users' visited websites even in complex, real-world conditions. The CountMamba performs robustly across three key scenarios: robust attacks (resisting various defenses), early-stage attacks (working with partial traffic), and multi-tab attacks (handling concurrent browsing). Unlike traditional WF attacks that depend on full traffic traces and are limited to single-tab browsing, CountMamba combines Windowed Traffic Counting Matrix (WTCM) and State-Space-Oriented (SSO) Classifier. WTCM is a coarse-grained, time-window-based representation of traffic that counts packets based on direction, timing, and cell relationships. This method increases resilience against defenses that alter packet-level patterns. SSO is a fine-grained, causal, and iterative classifier that updates predictions as new traffic data arrives. This enables accurate early-stage identification without needing full traffic traces and provides insights into user activity across multiple tabs.

B. Survey on WF Defenses

Tor encrypts and routes traffic through a network of relays to obscure user identity and destination, but it does not conceal traffic bursts or timing information, leaving it vulnerable to WF attacks. These attacks are particularly concerning in Tor, where users rely on strong privacy guarantees [160]. WF attacks against the first-hop (entry) relay can accurately infer user activity, especially in closed-world settings where the set of possible websites is predefined. To address this threat, researchers have proposed various WF defenses aimed at reducing the distinguishability of traffic patterns across websites, thereby diminishing the effectiveness of WF classifiers.

1) *Adaptive Padding-Based Defenses:* Adaptive traffic padding is a technique used to protect against WF attacks by dynamically injecting dummy traffic and introducing controlled delays into network traffic streams [53]. This dynamic behavior can be achieved by traffic obfuscation or randomized padding. Traffic obfuscation involves inserting dummy packets into statistically improbable timing gaps or low-activity regions of the traffic stream, making it more difficult for an attacker to identify meaningful patterns. Randomized padding

TABLE X: The comparison of representative adaptive padding-based defenses.

Defense Model	Effectiveness			Strengths	Drawbacks
	WF Attack Model	Closed World Acc (%)	Open World F1 (%)		
Camouflage [4]	Pa-SVM [4]	54.00→3.00	NA	1) No changes to anonymization protocols required; 2) Obfuscation through parallel page loading.	1) High bandwidth overhead; 2) In closed-world assumption for all web pages.
WTF-PAD [16]	K-NN [9]	91.00→17.25	95.00→50.00	1) Adaptive padding based on traffic bursts; 2) Lightweight protection with zero communication latency; 3) Low overhead and deployed in real-world Tor settings.	1) Less effective against deep learning-based and adaptive attacks; 2) Less effective in open-world scenarios; 3) Zero-latency but approximate high bandwidth overhead; 4) Do not significantly alter the burst patterns.
	Pa-SVM [4]	55.00→15.33	NA		
	DL-SVM [7]	83.70→23.00	NA		
	VNG++ [58]	98.00→26.00	NA		
FRONT [57]	K-NN [9]	NA	86.00→1.60	1) High effectiveness with lightweight bandwidth use; 2) Effective obfuscation of trace fronts; 3) Zero latency overhead; 4) Randomizes dummy packet at the beginning of a page load which will reduce the latency overhead.	1) Overhead is dataset dependent; 2) Require timestamp information from original traffic samples.
	CUMUL [11]	NA	76.00→13.00		
	K-FP [36]	NA	93.00→46.00		
	DF [93]	NA	94.00→40.00		
RBP [161]	CUMUL [11]	92.90→21.40	96.50→67.90	1) Low time delay; 2) Randomized bidirectional padding.	Higher bandwidth overhead than some lightweight methods.
	TF [95]	92.20→28.80	92.10→36.10		
	DF [93]	98.30→38.60	96.70→40.90		
RanDePad [78]	k-FP [36]	89.98→54.17	78.00→31.00	1) Low-latency; 2) Controllable overhead.	1) Feature leakage from the timing trace remains; 2) Low-efficiency in DL-based WF attack models.
	CUMUL [11]	90.77→50.39	90.00→34.00		
	DF [93]	94.57→62.40	91.00→29.00		
BRO [104]	DF [93]	72.20→43.00	NA	1) Lightweight and zero latency; 2) Beta distribution-based padding and conceals critical features; 3) Higher randomness and flexibility.	1) Limited scope to packet injection; 2) Low-efficiency defense against DL-based WF attack models.
	k-FP [36]	61.30→39.50	NA		
	CUMUL [11]	62.50→51.90	NA		

further enhances this effect by varying the timing and size of the inserted packets, thereby increasing uncertainty for WF classifiers. This adaptive approach enables the system to disrupt fingerprinting attacks while maintaining low overhead in terms of bandwidth and latency [77]. Fig. 7 illustrates the mechanism of adaptive padding. In the figure, coral rectangles represent outgoing packets, while light blue rectangles denote incoming packets. The light yellow and light green rectangles indicate the dummy packets added for outgoing and incoming traffic, respectively. The insertion of these dummy packets helps obscure burst patterns, timing information, and other features commonly exploited by WF attacks.

TABLE X summarizes existing representative adaptive padding-based defenses, highlighting their strengths and limitations to facilitate comparison for researchers. Juarez *et al.* [16] introduced WTF-PAD, an adaptive padding defense that protects Tor users from WF attacks. This defense employs adaptive padding to disrupt distinctive traffic patterns, thereby significantly reducing the accuracy of traditional ML-based WF attacks [4], [7], [9]. With zero latency overhead and less than 60% bandwidth overhead, WTF-PAD offers a more practical solution than previous defenses. Cui *et al.* [54] introduced an algorithm that generates realistic cover traffic to mitigate WF attacks. The proposed method uses a user's

historical network traffic to generate cover traffic that mimics their browsing habits, making it difficult for adversaries to distinguish real traffic from noise. Cui *et al.* [162] introduced an improved algorithm that also uses historical browsing data to generate realistic cover traffic, reducing attack accuracy to 14%–16% in both simulations and real-world experiments. Gong *et al.* [57] presented two lightweight defenses, FRONT and GLUE, which aim to protect against advanced WF attacks [9], [11], [36], [93] with minimal cost. FRONT obfuscates the beginning of traffic traces by injecting randomized dummy packets, while GLUE inserts dummy packets between traces to create the illusion of a single, continuous trace. Both methods operate without delaying client packets and require only a small number of dummy packets, offering a practical and efficient defense.

Pulls [55] introduced Interspace, a promising padding mechanism for Tor that offers a tunable balance between defense strength and bandwidth overhead in 2020. This work emphasizes the importance of probabilistic defenses and highlights the need for further research to overcome real-world deployment challenges, such as timing inaccuracies and adaptive attackers. Abusnaina *et al.* [67] proposed the Deep Fingerprinting Defender (DFD), a defense mechanism targeting deep learning-based WF attacks. DFD employs per-burst dummy packet injection in both one-way and two-way modes, disrupting traffic patterns inherent in Tor user traces. Luo *et al.* [161] introduced Random Bidirectional Padding (RBP) that combines time sampling with bidirectional dummy packet injection to obfuscate traffic features. RBP effectively disrupts inter-arrival time (IAT) features and packet distributions, making it more difficult for classifiers to extract meaningful patterns.

The protocol layer metadata (*e.g.*, cell sequences and timing) of Tor enables accurate classification of circuit types, such as onion service circuits versus regular clearnet circuits. To address this issue, Kadianakis *et al.* [68] presented the first comprehensive framework and adaptive padding strategies for defending against circuit fingerprinting in Tor. By exploring both delay-based and preemptive padding defenses, the authors demonstrate that practical, low-cost defenses can thwart powerful circuit fingerprinting adversaries. The work also opens new directions for integrating padding into Tor to protect against broader classes of traffic analysis attacks.

To reduce the requirement of high computation or additional infrastructure for previous WF defense works, Hong *et al.* [78] introduced Randomized Delaying and Padding (RanDePad), a WF defense mechanism designed to resist WF attacks while maintaining low latency and controllable bandwidth. RanDePad uses adaptive random delaying and adaptive random padding to disrupt the time distribution patterns of traffic. The key contribution is its ability to achieve a superior balance of low latency and controllable bandwidth compared to other state-of-the-art WF defenses, while effectively reducing the accuracy of DL-based attacks. Wang *et al.* [163] introduced Tree-structured Burst-sequence Padding (TBP), a defense mechanism that inserts dummy burst sequences to mimic fake resources. The method uses multiple random distributions to randomize both the timing and direction of traffic bursts, thereby obfuscating the user's online activity. This approach

obscures intra-class similarities and inter-class differences, reducing the true positive rate (TPR) of advanced CNN-based attacks such as Var-CNN to 44.7% with a 55% bandwidth overhead, outperforming defenses like FRONT by 15% in TPR reduction. Lu *et al.* [72] introduced WF Defense based on Distribution Distance Padding (WFD³P), a lightweight, zero-delay defense mechanism designed to combat WF attacks. WFD³P selects target traces from a pool with the maximum distributional distance from the user's traffic and injects dummy packets based on the selected trace's characteristics. Unlike global-pattern padding methods, WFD³P dynamically tailors padding to each source trace, improving defense granularity and effectiveness.

In 2023, Yang *et al.* [164] proposed Advanced Adaptive Padding (AAP), a zero-delay, moderate-overhead defense that extends the original Adaptive Padding (AP) approach. The AAP focuses on obfuscating HTTP burst patterns during website loading, providing enhanced protection for the Tor anonymity network. McGuan *et al.* [104] introduced Beta Randomized Obfuscation (BRO), a lightweight and zero-delay defense against WF attacks. BRO uses a beta distribution to strategically inject dummy packets, especially targeting the feature-rich beginning of traffic traces. With minimal bandwidth overhead (16.8–25.4%), BRO enhances trace randomness and dummy packet clustering, reducing the accuracy of state-of-the-art attacks like Deep Fingerprinting (DF) to 43%, compared to 54.2% under FRONT. Pulls *et al.* [165] introduced Maybenot, a flexible framework designed to implement traffic analysis defenses against metadata-based WF attacks in encrypted communication protocols like Tor, VPNs, and QUIC. Maybenot aims to provide a generalized, lightweight, and adaptable solution for deploying defenses without requiring extensive modifications to existing protocols. Huang and Du [166] proposed Break-Pad, which injects padding packets when the number of consecutive incoming packets exceeds a threshold, thereby breaking long bursts into smaller ones. In addition, Break-Pad uses randomized thresholds and padding counts sampled from probability distributions (Weibull for thresholds, Pareto for padding counts) to enhance unpredictability.

2) *Regularization-based defenses:* Even though adaptive padding-based mechanisms can efficiently defend against WF attacks by adding packet padding or dummy packets, attackers can still accurately estimate the accessed website using DL-based attack models through interval timing analysis and deep learning-based feature extraction. Because the padding scheme is often deterministic or follows predictable rules [108]. To defend against DL-based WF attacks effectively, regularization-based defenses intentionally reduce model performance on WF attacks by regulating the data representation, thereby diminishing the model's ability to learn discriminative features. As shown in Fig. 8, the regularization-based defense approach enforces fixed rules and patterns, such as uniform inter-packet intervals or consistent burst sizes, that all web traffic must follow [167]. TABLE XI summarizes representative regularization-based defenses.

In 2012, Dyer *et al.* [58] conducted the first comprehensive analysis of low-level countermeasures, such as per-

TABLE XI: The comparison of representative regularization-based defenses.

Defense Model	Effectiveness		Strengths	Drawbacks
	WF Attack Model	Closed World Acc (%)		
BuFLO [58]	Pa-SVM [4]	97.20±0.20→27.30±1.80	Resistant to fine-grained classifiers.	1) Extremely high bandwidth overhead and high latency; 2) Leaks coarse-grained features.
	P-NB [8]	98.20±0.90 →21.40±1.00		
	VNG++ [58]	93.90±0.30 →22.00±2.10		
CS-BuFLO [69]	Pa-SVM [4]	50.10→18.00	1) Congestion sensitivity; 2)Rate adaptation with limited leakage; 3) Adjustable overhead.	1) High bandwidth overhead; 2) Client-Server coordination required.
	DL-SVM [7]	75.10→<20.60		
DynaFlow [56]	K-NN [9]	88.00→6.00	1) No pre-established database of website traffic patterns required; 2) Real-time operation.	1) Impractical for real-world deployment.; 2) High bandwidth and latency overhead.
	k-FP [36]	94.30→18.40		
GLUE [57]	K-NN [9]	83.18→37.22	1) Obfuscates session boundaries that an attacker is impossible to obtain the beginning and end times of the trace.; 2) Resistant to the "Split Problem".	1) User behavior assumptions required; 2) Less effective if there is a long interval between two opening websites.
	CUMUL [11]	64.22→8.52		
	k-FP [36]	94.38→68.33		
	DF [93]	91.12→30.59		
RegulaTor [73]	Tik-Tok [12]	97.00→25.40	1) No packet content or full trace knowledge required; 2) Adjustable overhead; 3) Achieves the lowest attack accuracy among all methods with comparable overhead.	1) Requires parameter tuning and delaying data transmission; 2) Only focuses on regulating the coarse feature.
	DF [93]	98.40→19.60		
	CUMUL [11]	97.20→16.30		
Palette [19]	k-FP [36]	94.54→29.39	1) Provides strong protection against DL-based WF attacks; 2) Effective against adversarial training; 3) Less information leakage.	1) High bandwidth overhead; 2) Extra buffer required for each traffic regulation; 3) May cause a high traffic delay due to the traffic regulation.
	CUMUL [11]	94.81→10.96		
	DF [93]	98.23→20.27		
	Tik-Tok [12]	98.45→24.73		
	Var-CNN [96]	98.83→22.79		
	RF [119]	98.40→36.43		

packet padding, in the context of closed-world WF attacks. Their work introduced Buffered Fixed-Length Obfuscation (BuFLO), a defense that sends fixed-size packets at fixed intervals by adding dummy traffic, thereby obfuscating total traffic duration and bandwidth. BuFLO was shown to reduce WF attack [4], [8], [66], [106] accuracy by up to 5%, but its overhead was considered impractically high for real-world deployment. To address the inefficiency of BuFLO, Cai *et al.* [69] introduced Congestion-Sensitive BuFLO (CS-BuFLO) that adds rate adaptation, congestion sensitivity, and prac-

tical padding control. It offers strong protection with real-world deployability, though it still incurs higher overhead than lightweight defenses. In the same year, Cai *et al.* [15] introduced another defense method, Tamaraw, to achieve a better security/bandwidth trade-off than BuFLO. This method, combined with a theoretical framework and feature-based testing, offers a blueprint for building future defenses that are efficient and attack-agnostic. But the Tamaraw can only achieve non-uniform accuracy.

Later, Lu *et al.* [56] introduced DynaFlow, a WF defense

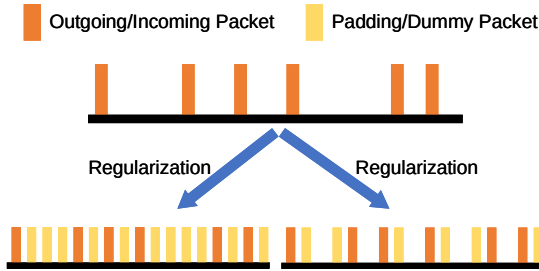


Fig. 8: An overview of the regularization-based defense mechanism. It adds dummy cells/padding to the incoming and outgoing traffic with a regularization.

mechanism by dynamically adjusting website traffic flows using the average inter-packet arrival time from the previous period. DynaFlow enhances security while improving efficiency by over 40% compared to previous WF defense mechanisms [15], [58], [69], eliminating the need for a pre-established database and extending protection to dynamically-generated websites. Experimental evaluations demonstrate that DynaFlow significantly reduces attacker accuracy while maintaining acceptable latency and bandwidth overheads, making it a promising and practical solution for safeguarding user privacy amid increasing surveillance. To address the challenges of high bandwidth and latency overhead, as well as the reliance on additional infrastructure, Holland *et al.* [73] proposed RegularTor. RegularTor mitigates these attacks by leveraging common patterns in typical web browsing traffic, thereby reducing both the resource overhead of the defense and the accuracy of state-of-the-art WF classifiers. Experimental results demonstrate that RegularTor reduces the accuracy of the Tik-Tok attack from 66% to 25.4%, significantly outperforming prior defense methods. To effectively defend against deep learning-based WF attacks with lower overhead, Chen *et al.* [168] proposed Timing-Assistant Traffic Burst Shaping (BM). This approach addresses the limitations of existing defenses, such as WTF-PAD and Walkie-Talkie, by incorporating temporal features. BM redefines traffic bursts using dynamic time thresholds and merges sequences from sensitive and non-sensitive websites into a unified “super-sequence” to obscure burst-level patterns. Later, Shen *et al.* [19] presented Palette, a practical defense against WF attacks that threaten user privacy in Tor. Unlike traditional defenses, Palette focuses on traffic cluster anonymization by grouping websites with similar traffic patterns into clusters and regulating their traffic to conform to a uniform pattern. This approach makes it significantly harder for attackers to distinguish between websites within the same cluster, effectively mitigating the risk posed by modern deep learning-based WF attacks.

3) *Traffic Morphing-Based Defenses*: Website morphing is another privacy-enhancing technique designed to defend against WF attacks. This mechanism will transform the observable traffic characteristics (*e.g.*, packet size distributions) of a webpage (the source) so it resembles another webpage (the target), thus misleading classifiers without excessive overhead [17], [106]. By adjusting specific traffic features, website morphing effectively shifts the classifier’s prediction away from the true class and toward a predefined, misleading target class. Fig. 9 illustrates the traffic morphing mechanism, which

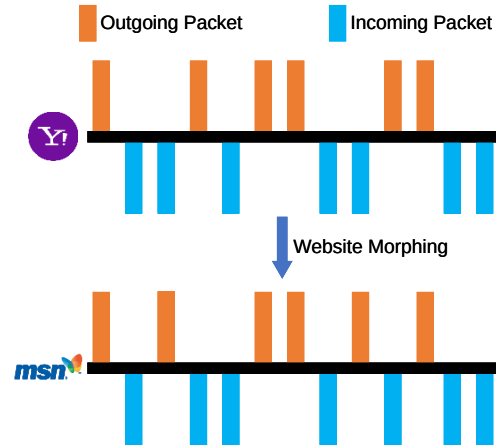


Fig. 9: An overview of the traffic morphing mechanism. It adds padding/dummy cells for both incoming and outgoing traffic and morphs the incoming or outgoing packets from one website to another website.

adds padding or dummy cells to both incoming and outgoing traffic, transforming the packet patterns to resemble those of a different website. TABLE XII shows a comprehensive comparison of the representative traffic morphing-based defenses.

In 2014, Supersequence [9] was introduced to transform the actual packet sequence of a website into a common supersequence shared across an anonymity set (*e.g.*, a group of packet sequences from different web pages). To implement this, the authors approximate the Shortest Common Supersequence (SCS) across packet sequences in an anonymity set. They cluster packet traces into sets that share similar characteristics and generate a supersequence for each set. When a page is visited, its trace is padded or reshaped to match the supersequence, preventing attackers from identifying the specific site visited. To address the problem of hampering the user experience and burdening the server with large overheads for previous works, Wang and Goldberg [45] presented Walkie-Talkie, which is designed to make sensitive and non-sensitive pages look the same. This approach employs half-duplex communication to group outgoing and incoming packets, along with burst molding to add fake packets, making it difficult for attackers to distinguish between sensitive and non-sensitive web page accesses. The authors demonstrate that Walkie-Talkie [45] achieves lower bandwidth and time overhead than existing defenses, effectively mitigating multi-page attacks. However, Walkie-Talkie requires prior knowledge of each site and is less effective for web pages with dynamic content.

Chan-Tin *et al.* [22] introduced a mechanism against WF attacks by leveraging traffic morphing and clustering to make websites within the same cluster indistinguishable. The proposed method groups websites based on their network traffic patterns (packet sizes and counts) and morphs all traffic in a cluster to resemble the largest website in that cluster, significantly reducing the attacker’s ability to identify specific sites. Al-Naami *et al.* [23] introduced BiMorphing, a technique that utilizes bi-directional burst sampling and optimization methods to obfuscate traffic patterns. This approach achieves minimal bandwidth overhead and zero-delay transmission. The defense is evaluated to outperform previous methods in both closed-

TABLE XII: The comparison of representative traffic morphing-based defenses.

Defense Model	Effectiveness			Strengths	Drawbacks
	WF Attack Model	Closed World Acc (%)	Open World Acc (%)		
Supersequence [9]	KNN [9]	91.00→6.80	NA	Optimal classification strategy.	High computational complexity of SCS.
Walkie-Talkie [45]	Pa-SVM [4]	81.00→44.00	NA	1) Efficient decoy selection and storage; 2) Low overhead and half-duplex communication simplifies traffic structure.	1) Requires decoy page priori knowledge about each site; 2) Cannot protect dynamic Content Well; 3) Webpage loading mechanism needs to be modified for the browser and deployment is challenging.
	DL-SVM [7]	94.00→19.00	NA		
	OSAD-SVM [35]	97.00→25.00	NA		
	K-NN [9]	95.00→28.00	NA		
	CUMUL [11]	64.00→20.00	NA		
	k-FP [36]	86.00→41.00	NA		
BiMorphing [23]	BIND [169]	80.04→15.57	98.96→86.35	1) Bi-directional burst morphing; 2) Lower Bandwidth Overhead and zero-delay transmission.	1) Computational overhead for initialization phase; 2) Sensitivity to target selection; 3) Limited in defending against DL-based WF attacks.
	CUMUL [11]	91.02→19.64	96.50→85.06		
	K-NN [9]	83.85→12.93	NA		
WFGuard [24]	DF [93]	98.35→14.18	96.80→15.52	1) Innovative use of fuzzing and neuron feedback; 2) Minimal overhead with high defense performance.	1) Prior knowledge of website required; 2) Still vulnerable to adversarial training.
	Var-CNN [96]	98.40→11.04	97.23→11.22		

world and open-world scenarios. Previous defenses [23], [45], [56], [69] often struggle to counter DL-based WF attacks effectively without introducing substantial bandwidth or latency overhead. To solve this issue, Ling *et al.* [24] presented WFGuard, a traffic morphing defense technique against DL-based WF attacks based on fuzzing testing. WFGuard leverages fine-grained neuron information from WF classifiers to design a joint optimization function, which is then optimized using gradient ascent to simultaneously maximize both neuron values and the likelihood of misclassification in DL-based WF classifiers. During each traffic mutation cycle, a gradient-based approach is employed to generate dummy traffic injection patterns, continuously evolving the traffic until a pattern is found that successfully deceives the classifier. Extensive evaluations show that WFGuard significantly reduces the accuracy of DL-based WF classifiers (*e.g.*, DF and Var-CNN) to as low as 4.43%, while only incurring a modest 11.04% bandwidth overhead.

4) *Adversarial Perturbation-Based defenses*: In recent years, DL-based WF attacks represent a significant evolution in the ability to de-anonymize users based on their encrypted network traffic. By automating the feature extraction process and leveraging powerful models like CNNs, RNNs, and LSTMs, these attacks can achieve very high accuracy in identifying websites, even under challenging conditions like traffic obfuscation [173], [174]. Over the years, adversarial

perturbations, derived from the field of adversarial ML/DL, refer to carefully designed modifications of input data that cause ML/DL models to make incorrect predictions [175], [176]. Fig. 10 presents an overview of the adversarial perturbation-based mechanism. The adversarial perturbation-based mechanism adds minimal, often imperceptible, noise or modifications to a legitimate input and thereby forces a target classifier to produce an incorrect prediction. This subsection will explore adversarial perturbation-based defenses for WF, comparing their methods, effectiveness, strengths, and drawbacks. The comparison of the representative adversarial perturbation-based defenses is summarized in TABLE XIII. TABLE XIII presents the effectiveness of defenses, showing changes in WF attack accuracy (attack rate) and defense success rate (defense rate). The strengths and drawbacks of each defense model are also summarized in TABLE XIII.

Previous defenses like traffic morphing [45] and adaptive padding [58], [69] are limited by static patterns and high detectability. To solve this issue, Imani *et al.* [177] presented an Adversarial Traces defense method by adding padding to a Tor traffic trace in a manner that reliably fools the classifier into classifying it as coming from a different site. Li *et al.* [109] addressed the challenge of evading internet censorship through traffic analysis attacks, such as WF and protocol fingerprinting (PF). The authors propose FlowGAN, which uses Generative Adversarial Networks (GANs) to dynamically

TABLE XIII: The comparison of representative adversarial perturbation-based defenses.

Defense Model	Effectiveness			Strengths	Drawbacks
	WF Attack Model	Attack Rate Acc (%)	Defense Rate (%)		
Mockingbird [60]	DF [93]	98.00→38.00	>80.00	1) Protection against multiple-round intersection attacks; 2) Randomized target perturbation strategy; 3) Less information leakage.	1) Deployment complexity; 2) Requires prior knowledge of the complete traffic burst patterns.
	Var-CNN [96]	99.00→30.00	>80.00		
	CUMUL [11]	92.00→20.00	>80.00		
	k-FP [36]	92.00→26.00	>80.00		
	K-NN [9]	90.00→12.00	>80.00		
PST [170]	DF [93]	91.00→44.00	>95.00	1) High evasion rate and low overhead efficiency; 2) Only a few past bursts of a trace are needed.	1) Vulnerability to adversarial training; 2) Time consuming in searching fuzzy adversarial perturbation direction; 3) Only modifies the future part of the traffic trace, limiting the defense's effectiveness.
	CNN [94]	83.86→32.39	>95.00		
	SDAE [94]	82.64→39.08	>90.00		
	LSTM [94]	80.41→31.26	>90.00		
	CUMUL [11]	64.00→20.00	NA		
BLANKET [61]	DF [93]	92.00→1.00	NA	1) Works in real-time on live traffic; 2) High transferability and flexibility.	1) Vulnerability to defense-aware retraining; 2) Potential to informative features leakage.
	Var-CNN [96]	93.00→1.40	NA		
WF-UAP [21]	DF [93]	97.50→16.80	73.40	1) Prior knowledge of traffic patterns is not required; 2) Defense against adversarial training.	1) Fixed-length UAP matching may reduce precision in real-world traces; 2) Improperly tuning may increase bandwidth overhead with loss-tuning-based bandwidth control.
	Var-CNN [96]	98.30→15.50	82.30		
	AWF [94]	94.10→12.40	85.50		
	CUMUL [11]	92.70→10.60	NA		
	k-FP [36]	92.20→9.20	NA		
	K-NN [9]	90.10→8.20	NA		
Minipatch [74]	AWF [94]	NA	>99.70	1) Real-time and real-world deployment; 2) Lower bandwidth overhead with few dummy packets injection.	1) Unable to withstand attacks that involve retraining the classifier; 2) Requires prior knowledge of visited website.
	DF [93]	NA	>97.30		
	Var-CNN [96]	NA	>99.20		
Acup3 [171]	DF [93]	97.00→17.86	NA	1) Robust against adversarial training; 2) Universal and fast perturbation generation; 3) Efficient and lightweight.	1) Trade-off tuning required; 2) No real-time dynamic adaptation; 3) Potential for overhead increase with more data.
	Var-CNN [96]	98.00→24.29	NA		
	AWF [94]	90.96→15.07	NA		

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Table XIII: Continued from previous page.

Defense Model	Effectiveness			Strengths	Drawbacks
	WF Attack Model	Attack Rate Acc (%)	Defense Rate (%)		
Surakav [46]	k-FP [36]	73.62→1.00	NA	1) No prior knowledge required; 2) Defend against adversarial training by random response; 3) Can mimic realistic traffic patterns of different webpages by GAN.	1) High bandwidth and latency overhead, and not cost-effective for real-world use; 2) Feature leakage still possible.
	CUMUL [11]	74.23→2.74	NA		
	DF [93]	96.24→8.14	NA		
	Tik-Tok [12]	96.68→62.8	NA		
ALERT [47]	DF [93]	97.05→12.68	>61.48	1) Low communication overhead; 2) GAN-based universal perturbation generator and pluggable transport compatible.	1) Defense efficiency depends on substitute classifier; 2) Potential to informative features leakage.
	AWF [94]	87.74→8.29	>81.84		
	Var-CNN [96]	95.88→7.49	>76.79		
KimeraPAD [172]	Tik-Tok [12]	96.17→20.92	NA	Low bandwidth overhead and real-time implementation.	1) Requires prior website knowledge; 2) Performance depends on surrogate model quality; 3) Sensitive to network condition.
	RF [119]	94.44→22.72	NA		
	TF [95]	91.10→18.53	91.70		
	DF [93]	91.90→18.48	81.50		
	CUMUL [11]	95.42→0.55	95.50		
	k-FP [36]	97.44→8.79	NA		

camouflage censored traffic as benign network flows, thereby bypassing censorship. Cai *et al.* [178] introduced JumpEstimate, a black-box countermeasure against WF attacks, which leverages adversarial machine learning to confuse the decision boundaries of DL and ML-based WF classifiers. Without relying on manually extracted features, JumpEstimate uses adversarial traffic generated through an improved boundary attack and Monte Carlo estimation to maximize confusion while resisting retraining by the attacker.

Deep neural networks are known to be vulnerable to adversarial examples. Hou *et al.* [59] introduced WF-GAN, which utilizes adversarial learning techniques to generate adversarial examples for WF classifiers. The authors demonstrate that WF-GAN effectively generates adversarial examples to obscure traffic patterns, achieving a 90% success rate with minimal overhead (5-15%) in both untargeted and targeted defense scenarios. The results indicate that WF-GAN outperforms existing defenses, such as Walkie-Talkie, providing a robust solution for protecting encrypted traffic against sophisticated WF attacks. Rahman *et al.* [60] introduced Mockingbird, a technique designed to generate traces that resist adversarial training by introducing randomness into the process. Instead of following predictable gradients, Mockingbird explores the space of viable traces through random movements, making it more challenging for adversarially trained classifiers to defend against. Unlike conventional adversarial techniques,

which are typically vulnerable to adversarial retraining, Mockingbird evades such defenses by perturbing network traces in a way that progressively shifts the source trace toward multiple randomly selected target traces. This approach does not rely on gradient-based optimization, complicating adversarial adaptation and defense. To address the limitation of requiring the entire original traffic trace for adversarial perturbations in WF defenses, Jiang *et al.* [170] introduced PST, a practical adversarial learning-based defense against WF attacks, which dynamically generates perturbations using only the initial bursts of network traffic rather than requiring the entire trace. PST employs an encoder-decoder neural network with attention mechanisms to predict subsequent traffic bursts, then applies a modified Boundary Attack to compute adversarial perturbations, and transfers these perturbations to the remaining traffic. Evaluation results show that PST effectively disrupts the network traffic pattern, achieving a high evasion rate of 87.6%. This outperforms Walkie-Talkie by over 31.59% at the same bandwidth overhead, while only requiring observation of 10 transferred bursts.

Later, Nasr *et al.* [61] introduced an adversarial approach to counter deep neural network (DNN)-based traffic analysis techniques. The key idea behind their adversarial perturbations algorithm is to generate “blind” perturbations, which are independent of the target inputs, by solving specific optimization problems. This method operates in real-time, without requiring

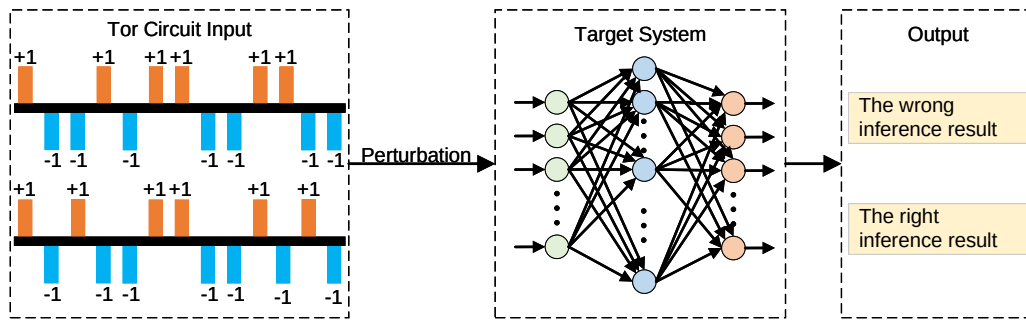


Fig. 10: An overview of the adversarial perturbation-based mechanism. It adds minimal noise or perturbations to the input, then forces the attacker’s classifier to misclassify the perturbed input.

prior knowledge of network traffic, making it highly practical. It perturbs live network traffic on-the-fly, without the need to buffer packets or rely on any prior understanding of traffic patterns. Building on Nasr’s work, Shan *et al.* [179] proposed Dolos, a defense against WF attacks that can be applied in real-time to network traffic. Dolos uses parameterized randomness to thwart adaptive attacks, including adversarial training. A key element of Dolos is the use of trace-agnostic adversarial patches, generated via a user-side secret, to protect network traces. Unlike adversarial examples tailored for specific inputs, these patches are pre-computed sequences of dummy packets that can induce misclassifications across a broad range of inputs, effectively safeguarding visits to specific websites. Hou *et al.* [20] proposed a universal adversarial defense against WF attacks by generating perturbations that deceive DNN-based classifiers, without requiring prior knowledge of the full traffic trace. The method utilizes the Jacobian matrix to estimate the effect of dummy packet insertions on classifier outputs, facilitating efficient perturbation selection. In the same year, Hou *et al.* [180] introduced another defense mechanism, Attack to Attack (A2A), which uses adversarial examples to disrupt the attacker’s classifier. A2A operates as a proxy near the client, perturbing encrypted traffic by iteratively inserting dummy packets based on feedback from a substitute model trained to mimic the attacker’s classifier. Evaluated on a public Tor dataset [93], [94], A2A reduces the attacker’s accuracy to 57.6% with only 2.2% bandwidth overhead, significantly outperforming existing defenses like WTF-PAD and Walkie-Talkie, which incur higher overheads (31–64%). Wang *et al.* [181] introduced Local Optimal Mutation (LOM), a method that mutates sensitive traffic traces toward target populations of non-sensitive traces using a gradient-descent-inspired strategy. This approach minimizes computational costs while maintaining effectiveness. By combining the benefits of adversarial samples with a WF defense method based on local optimal mutation, LOM achieves the generation of adversarial traffic traces with shorter preparation times.

Inspired by the universal adversarial Perturbations (UAPs) mechanism [175], Sun *et al.* [21] developed a GAN-based approach for UAP generation, called WF-UAP. Leveraging the power of GANs, this method efficiently creates UAPs that make adversarial traces harder to distinguish from original traffic. The crafted UAPs are independent of target network traffic traces and can disrupt the classifiers used in WF attacks when applied to any traffic trace. In addition, this

method can inject dummy packets into live traffic, making adversarial traces indistinguishable from real ones while minimizing bandwidth overhead. Adversarial perturbations have excellent defense capabilities against WF and low bandwidth overhead, but they need to get the complete website traffic to generate defense data, which is obviously impractical. Tang *et al.* [182] proposed segmented adversary defense (SAD) for deep learning-based WF attacks. In SAD, sequence data are divided into multiple segments to ensure that SAD is feasible in real scenarios. The adversarial examples for each segment of data can be generated by SAD. Finally, dummy packets are inserted after each segment original data.

Gong *et al.* [46] proposed a solution using Generative Adversarial Networks (GANs) to generate realistic traffic patterns that mimic the behavior of various web pages. These synthetic traces are used to regulate packet transmission, obscuring the website being visited by introducing randomness in both packet sizes and sending times. This dynamic adjustment of packet bursts based on buffered data makes it significantly harder for attackers to distinguish between different websites. Jiang *et al.* [183] introduced Targeted LightwEight Defense (TED), a lightweight and efficient method to defend against WF attacks in the Tor network. TED introduces trace-level defenses, customizing the defense for each unique trace—even for the same website. In addition, TED leverages a fully convolutional network (FCN) combined with adapted Grad-CAM to extract key discriminative features from traffic traces. These are the elements deep learning-based WF attacks rely on. TED then precisely perturbs these key features by injecting a small number of dummy packets, misleading the attacker’s classifier without heavy overhead.

Many previous adversarial perturbation-based WF defenses have been criticized for relying on unrealistic assumptions, such as full access to the target model or needing to operate on the entire website trace. To overcome these limitations, Li *et al.* [74] introduced Minipatch that strategically injects few dummy packets in real-time traffic to evade the attacker’s classifier. Unlike previous adversarial-based defenses, which suffer from high bandwidth overhead or unrealistic assumptions (*e.g.*, requiring white-box access to the attack model), Minipatch operates in a black-box setting and maintains low overhead (less than 5%). Bai *et al.* [184] introduced WGAN-GA that combines Wasserstein Generative Adversarial Networks (WGAN) with Genetic Algorithms (GA) to generate adversarial traffic samples that obfuscate fingerprinting attempts. Hol-

land *et al.* [185] employed a Generative Adversarial Network (GAN)-inspired architecture to ensure the defense adapts to evolving attacker strategies. Yin *et al.* [186] introduced Greedy Injection Attack (GIA) and Sniper, two adversarial example-based WFD frameworks. These methods dynamically perturb network traffic to mislead DL-based WFAs with minimal bandwidth overhead. Prior WF defenses often fail to enforce real-time constraints or require symmetric proxy deployments. To address this issue, Li *et al.* [187] exploited patterns in packet timing and size to compromise user privacy. Then, the authors proposed Prism, a lightweight, asymmetric framework that dynamically perturbs live traffic to evade detection while maintaining transmission integrity. Qiao *et al.* [171] proposed a black-box WF defense called Acup3, which operates under the assumption that the attacker's classifier is inaccessible and unknown. This technique makes traffic traces from different websites appear more similar, increasing the difficulty of classification. Acup3 generates perturbations without needing access to the actual traffic traces, making it practical for real-world deployment. By diversifying the perturbations applied to traffic traces of different users visiting the same website, Acup3 reduces the effectiveness of AT, as the classifier cannot easily generalize from previously seen perturbations.

To mitigate the high bandwidth and latency overhead of prior WF defenses, Jiang *et al.* [172] introduced KimeraPad that uses deep reinforcement learning (DRL) to generate adversarial perturbations in real-time, making it difficult for attackers to classify traffic patterns accurately. KimeraPAD can be implemented in real-time without requiring prior knowledge of the exact packet sequences or their lengths, making it practical for real-world use. Wang *et al.* [188] proposed CMAES-WFD, a black-box adversarial defense against WF attacks, which leverages the Covariance Matrix Adaptation Evolution Strategy (CMAES) to generate perturbations that mislead deep neural network (DNN)-based classifiers while minimizing bandwidth overhead. The method transforms perturbation generation into a constrained optimization problem, injecting dummy packets into critical positions of traffic bursts to disrupt classification without requiring access to the target model's internal parameters.

Existing defenses, particularly those based on Adversarial Examples (AEs) [60], [189], face challenges such as requiring complete traffic traces for perturbation calculation and being vulnerable to Adversarial Training (AT) by attackers. To address this issue, Qiao *et al.* [47] presented a trace-Agnostic LightwEight Transferable Robust (ALERT) WF defense to resist DNN-based WF attacks. ALERT uses a GAN-based generator to create universal perturbations without requiring access to real-time traffic traces, enabling packet manipulation before browsing starts. It incorporates a random website imitation strategy, where each user's traffic is perturbed to mimic a randomly selected website, introducing high diversity and mitigating adversarial training (AT) effectiveness. In closed-world experiments, ALERT reduces attack success rates (ASRs) below 13% for strong WF attacks like DF. In open-world scenarios, ALERT achieves the best trade-off between true and false positive rates while maintaining minimal communication overhead (19.9%). Jiang *et al.* [48] introduced RUDOLF,

an efficient and adaptive WF defense based on Soft Actor-Critic (SAC) reinforcement learning. RUDOLF dynamically adds dummy packets in real-time by training an RL agent to make burst-level decisions, eliminating the need for complete traffic traces and making it practical and adaptive. The agent receives feedback via a dual-component reward function that balances evasion success against bandwidth overhead, allowing it to learn optimal perturbation strategies. Evaluations show RUDOLF achieves attack accuracies as low as 15–20% with less than 30% bandwidth overhead.

5) *Other WF defense methods:* Besides the four WF defense methods discussed above, some researchers have proposed alternative approaches targeting different layers of the Onion routing protocol as well as distinct characteristics of traffic flows.

Cherubin *et al.* [190] proposed two application-layer defenses: ALPaCA, a server-side defense that modifies webpage content to obscure traffic patterns, and LLaMA, a lightweight client-side defense that randomizes request orders and introduces delays. The proposed defenses significantly reduce the accuracy of WF attacks, demonstrating their potential effectiveness in enhancing privacy for users of anonymous web services. De *et al.* [114] proposed two lightweight defenses, TrafficSliver-Net and TrafficSliver-App, which leverage traffic splitting over multiple Tor entry nodes to limit the data observed by any single entry node and distort repeatable traffic patterns exploited by WF attacks. The proposed methods aim to infer the content of encrypted and anonymized connections by analyzing patterns in data flows. Jahani *et al.* [191] proposed a WF defense model by minimizing the autocorrelation property in network traffic instances, which is identified as a key factor enabling WF attacks.

Henri *et al.* [192] proposed HyWF, which splits traffic between two direction networks using a random splitting strategy. For each website access, a random probability determines how packets are distributed between the networks. This randomness makes it difficult for an adversary to reconstruct the original traffic pattern. Later, Reuter *et al.* [193] presented Traffic Splitting for Tor, a defense against WF attacks that leverages multipath transmissions to obscure traffic patterns and thwart malicious entry nodes. The approach distributes Tor cells across multiple entry relays, disrupting identifiable timing and volume patterns while minimizing overhead.

Zhang *et al.* [194] introduced Unsupervised and Adaptive Tor Website Fingerprinting (UAF), which is designed to address the challenge of domain shifts in Tor WF attacks. Traditional WF methods assume that training and testing data share similar distributions, which is unrealistic due to diverse user environments. UAF leverages multi-source domain adaptation (MDA) to align features between labeled source domains and unlabeled target domains without requiring target labels. It employs three trace representations (raw directions, raw timestamps, and directional timestamps) to retain discriminative information and combines source-specific classifiers based on trace length distributions.

Ling *et al.* [195] proposed a genetic programming approach to search for effective dummy cell injection patterns that can mislead DL-based WF classifiers. The method involves

randomly injecting dummy Tor cells into labeled traffic traces and using a fitness function to evaluate the effectiveness of the generated variant traces. A fitness function based on feature distance is designed to guide the search for successful variant traces. The function calculates the distance between the feature vector of a variant trace and the average feature vector of a target website. A mutation direction control mechanism is introduced to ensure efficient search by using a sliding window to monitor the progress of the search.

The splitting of traffic in the heterogeneous network resources has emerged as a new defense mechanism to disrupt traffic characteristics. But the previous works are inefficient and costly. To address this challenge, Wang *et al.* [196] introduced PWR (Path Weighted Random), a multipath traffic splitting mechanism designed to counter WF attacks by leveraging heterogeneous network resources. PWR can dynamically distribute traffic across multiple paths based on real-time link conditions, such as delay, packet loss, and congestion, using a Dirichlet distribution to randomize path selection. This approach disrupts traffic patterns, making it harder for attackers to classify user behavior, while maintaining transmission efficiency through ordered packet scheduling and link redundancy padding.

Gu *et al.* [197] presented an innovative and practical solution for defending against WF attacks, showcasing the potential of adversarial techniques in enhancing online privacy without incurring prohibitive costs or overhead. This research contributes significantly to the field of network security, specifically in protecting user privacy while browsing the web. Liu *et al.* [18] presented SMART, a lightweight defense that splits traffic across multiple Tor entry relays, disrupting statistical features used by WFAs. SMART offers a practical, efficient defense against WFAs by combining multi-path routing and redundancy. However, Tor enables anonymous communication but remains vulnerable to flow correlation attacks that match traffic patterns between its entry and exit points. MUF-FLER [198] addresses these issues by dynamically reshaping egress traffic through real-time connection mapping without adding padding or delays. It effectively disrupts correlation attacks, achieving only 1% true positive rate at 1% false positive rate, with minimal bandwidth and lower latency—while maintaining compatibility with the existing Tor infrastructure.

VI. FUTURE OPPORTUNITIES AND CHALLENGES

In this section, we will discuss the challenges and open issues in the dataset, attacks, and defenses, highlighting our findings and specific considerations in literature reviews.

A. The challenges of dataset collection and construction

High-quality Tor traffic datasets are vital for real-world website fingerprinting analysis, as they help train accurate and robust machine learning and deep learning classifiers. However, these datasets often rely on assumptions that don't hold up in real-world scenarios.

1) *Webpage assumption*: The webpage assumptions mainly refer to the types of webpages that users browse. These include the single-page assumption, static webpage, passive webpage, cache-disabled assumption, replicability, traffic parsing, and multi-tab assumption with a fixed overlap ratio [27]. These assumptions simplify the scenario, or overestimating the adversary's capabilities gives the attacker an unrealistic advantage.

Single-page assumption. The single-page assumption refers to the common assumption by researchers that the observer monitors only a single page on a website, typically the homepage. Some researchers believe that monitoring just the homepage is sufficient to determine whether a user has visited the site. However, this assumption is considered unrealistic [35], as users are likely to browse both the homepage and internal pages of a website.

Static webpage. The static webpage assumption suggests avoiding WF attacks on sites where content changes frequently, such as news websites or video content platforms that update every few minutes [35]. Training a traffic classifier is often time-consuming, and when webpage content changes, the attacker must retrain the classifier using new browsing traffic, which is very costly. As a result, attacks typically target static webpages or websites with regularly updated content, such as government agency sites.

Passive webpage. The passive webpage assumption refers to the idea that when Tor users browse a webpage, all active content related to the page is either disabled or inactive. Aminuddin *et al.* [27] defined active content as traffic traces resulting from dynamically loaded elements such as Flash, Java Applets, or JavaScript. Additionally, ads, plugins, and special modifications to the homepage during holidays should also be disabled, as such dynamic content can cause the traffic pattern of the same webpage to differ with each load.

Cache-disabled assumption. Researchers assume that Tor users disable browser caching so that every time a webpage is visited, the browser downloads all of its resource files, avoiding access to any local cache. This ensures that repeated visits generate the same traffic patterns. While the Tor Browser is designed to clear cached data between sessions to enhance privacy, it still makes use of cache files during the same browsing session. As a result, subsequent visits within a session may load certain assets from the cache instead of re-downloading them, thereby altering the observed traffic patterns. This caching effect introduces variability into network traces, potentially reducing the reliability of dataset assumptions and challenging the validity of experimental results that overlook this factor.

Replicability. Replicability in website fingerprinting research assumes that an attacker can reproduce the Tor user's system and network environment during traffic simulation, including factors such as browser version, operating system, network bandwidth, and latency conditions. This assumption simplifies experimental design, as it allows researchers to generate training data under controlled conditions that are presumed to match those of the target user. However, achieving such replication in real-world scenarios is highly challenging. User environments are inherently diverse and dynamic, varying not only across individuals but also across time due to software

updates, fluctuating network conditions, and hardware differences. In particular, for multi-victim targeting scenarios, the attacker would need to replicate multiple heterogeneous environments simultaneously, which may be infeasible in practice. As a result, datasets and models trained under strict replicability assumptions risk overestimating attack performance, as they fail to capture the variability and unpredictability of real-world Tor usage. Recognizing and addressing these limitations is therefore essential for developing realistic datasets and evaluating the robustness of fingerprinting attacks and defenses.

Traffic Parsing. Traffic parsing assumes that an attacker can accurately determine the start and end of a webpage browsing session within the Tor traffic stream. In reality, Tor traffic is transmitted in uniform fixed-size cells, and browsing traces may be intertwined with unrelated background activities or overlapped with traffic from other webpages visited in close succession. These factors obscure session boundaries and hinder precise segmentation, preventing researchers from directly applying trained classifiers to the entire captured traffic. Therefore, effective website fingerprinting in practice requires robust parsing techniques or models that can operate reliably under noisy and overlapping traffic conditions.

Fixed Overlap Ratio in Multi-tab assumption. To simulate multi-tab browsing, researchers often assume that webpage traffic overlaps at fixed ratios such as 10%, 20%, or 50%. While this controlled setup simplifies experimentation, it does not accurately reflect real-world conditions. In practice, users open, close, and switch between tabs in highly unpredictable ways, leading to dynamic and irregular overlap of traffic flows. As a result, replicating realistic multi-tab behavior and determining precise overlap ratios in collected datasets remain significant challenges for website fingerprinting research.

2) *Concept drift:* Concept drift in WF attacks refers to the phenomenon where changes in webpage content over time alter the associated traffic patterns, thereby reducing the accuracy of classification models. Modern websites often update their content frequently, whether through dynamic elements, advertisements, multimedia, or layout modifications, all of which can significantly shift traffic characteristics. Training a traffic classifier is typically resource-intensive, requiring large volumes of labeled data and extensive computation.

Juarez *et al.* [6] are the first to reveal the impact of website content updates on WF attacks. They observed that attack accuracy dropped by 50% in less than 10 days and fell to nearly zero after 90 days. Attarian *et al.* [199] conducted a systematic survey of data stream mining algorithms for handling concept drift, comparing approaches such as Adaptive Hoeffding Tree [199], Concept-Adapting Very Fast Decision Tree (CVFDT) [200], OzaBag [201], and Extreme Fast Decision Tree (EFDT) [202]. Among these, the Adaptive Hoeffding Tree achieved the highest accuracy by automatically detecting drift and updating its parameters. Building on this, Attarian *et al.* [64] proposed the Adaptive Online WF Attack (AdaWFPA), which models WF using the Adaptive Hoeffding Tree and incrementally updates the classifier with each new training instance. This design enables continuous adaptation to evolving website versions and sustained performance over time. However, since the model is updated continuously,

it cannot be directly compared with the drift resilience of existing deep learning-based approaches.

Wang *et al.* [125] introduced snapshot WF (snWF), an ensemble method leveraging neural networks. Their snapshot ensemble strategy employs multiple snapshot models during testing, with final predictions obtained by averaging across models. The study further showed that concept drift has an even stronger impact in open-world scenarios, where performance degradation is more severe. Thus, evaluating drift in realistic open-world environments remains an important research challenge.

When concept drift occurs, attackers are forced to retrain classifiers on newly collected traces—a process that is both costly and time-consuming. Many WF attacks focus on relatively static websites or those with predictable, infrequent updates (*e.g.*, government portals) to maintain effectiveness. However, this assumption of stable targets highlights a critical limitation of WF attacks in dynamic, real-world environments. It underscores the need for adaptive models that can handle evolving traffic patterns without requiring complete retraining.

3) *User behavior assumption:* The effectiveness of WF attacks is often overstated due to flawed assumptions about user behavior. These assumptions, made from an attacker’s perspective, simplify the complex reality of internet use, leading to overly optimistic results in research. The primary assumptions include the closed-world assumption, the sequential browsing assumption, and the traffic isolation assumption.

The closed-world assumption dictates that a Tor user will only visit websites from a predetermined, finite set that an attacker has already collected. Researchers use this simplified scenario to train and test classifiers, which often leads to remarkably high accuracy rates. While useful for comparing the performance of different models, this controlled environment doesn’t reflect the open-ended nature of the real internet, where users can browse an infinite number of sites. Consequently, a classifier trained in a closed world is likely to perform poorly in a real-world setting where it encounters traffic from unknown websites [27].

The sequential browsing assumption simplifies user behavior by assuming that a user loads one webpage at a time and only moves to the next page after the previous one has fully loaded [37]. This eliminates any overlap in traffic from different websites. In reality, due to Tor’s slower connection speeds, users commonly open multiple browser tabs simultaneously. This concurrent browsing behavior generates a chaotic mix of overlapping traffic from different sites, making it incredibly difficult to isolate and identify the traffic of a single webpage.

The traffic isolation assumption presumes that a Tor user is only browsing websites and not performing any other online activities in the background, such as downloading files, streaming music, or updating their operating system. In practice, a user’s background network activities are a significant source of “noise” that can interfere with a website fingerprinting attack. The traffic generated by other applications can obscure the unique patterns of a webpage’s traffic, making it much harder for an attacker to identify the site.

4) *Network situation:* In WF attack and defense studies, researchers typically assume that the collected datasets

are captured under consistent network conditions. However, network congestion can introduce significant delays, which directly affect traffic patterns. WF models often rely on traffic features extracted within fixed time slots, as the exact start and end of a website trace are usually unknown. Prolonged delays increase the interval between consecutive packets and reduce the number of packets captured within the designated time slot. Consequently, the extracted features may no longer accurately represent the website’s traffic pattern, leading to degraded attack performance. This effect is particularly pronounced in dynamic or congested network environments, highlighting the need for WF models and defenses to account for variable latency and packet timing variability.

B. The challenges of WF attacks

1) *Dataset limitation*: Collecting fresh, diverse, and reliable datasets is a major challenge for website fingerprinting. Attackers need up-to-date data to maintain high accuracy, but this is a resource-intensive process. Consequently, resource-constrained attackers must compromise by either limiting the number of monitored websites, using outdated data, or reducing the number of training samples per site [26]. Two primary methods for dataset collection have emerged, each with its own trade-offs: the snowball method and the gold panning method [26].

The snowball method involves simulating a wide range of realistic user actions to generate and collect traffic traces. It begins with a small set of user behaviors and progressively expands to include different combinations of software, hardware, and environmental factors. The “snowball” name comes from the way the dataset grows. While this method can produce datasets with highly accurate ground truth, it requires significant manual effort and is not easily scalable.

The gold panning method involves passively collecting large volumes of traffic directly from real-world network devices, such as a campus network’s egress router. The resulting traffic is highly diverse and reflects real-world conditions. However, the process is akin to “panning for gold” because the target traffic is mixed with a massive amount of unrelated background data. This requires extensive, time-consuming filtering to isolate the relevant traces. Furthermore, a significant drawback is the difficulty of obtaining reliable ground truth for the vast majority of encrypted traffic collected this way.

2) *WF attack model building*: Tor traffic conceals the payload segments of packets, which significantly reduces the amount of information available for analysis. Consequently, most existing research relies on ML/DL techniques to build traffic analysis models. As mentioned earlier, traffic analysis is typically framed as either a supervised or an unsupervised classification problem, with classifiers generally falling into two categories.

The first category consists of traditional ML models, such as K-NN, SVM, and Decision Trees, which are carefully trained using empirically crafted features. The second category includes DL models, such as CNN, LSTM, and Transformer, which can automatically learn effective features directly from raw input data, thereby avoiding the complex and time-consuming process of manual feature engineering. A growing

trend combines deep learning and traditional machine learning by first extracting effective features with deep learning models and then using those features to train traditional classifiers. To further improve classifier performance, several challenging issues must be carefully addressed.

Effectiveness. To facilitate the classification or clustering of different classes of Tor traffic, an effective representation should be sufficiently discriminative to clearly separate each class from the others. Specifically, a strong representation should ensure that traffic instances are similar to those within the same class while remaining distinct from those in different classes. Existing representations typically abstract encrypted traffic from one or more aspects, such as packet order, timing, direction, and frequency.

Beyond these surface-level features, future research can explore the interaction processes between communication components—such as client-server exchanges—and enrich traffic data with more semantic information from the application layer. Such an approach can offer deeper insights into the fundamental reasons behind traffic differences across classes. Additionally, suitable evaluation metrics are required to quantitatively assess the effectiveness of various representations.

The fundamental goal of a classifier is to achieve high effectiveness, which can be measured using metrics such as accuracy, precision, and recall. Most existing studies evaluate classifier effectiveness in the closed-world setting [203], where traffic is assumed to originate from a limited number of targeted sources (*e.g.*, specific websites). However, a classifier that performs well in a closed-world scenario often loses effectiveness in the more realistic open-world setting, where encrypted traffic originates from a much larger number of untargeted sources. In this context, targeted traffic may represent only a small fraction of the total traffic, and the classifier must be capable of reliably distinguishing targeted traffic from untargeted traffic. Furthermore, existing defense methods negatively impact classifier effectiveness because fewer discriminative features can be extracted from protected traffic. A key challenge, therefore, is to investigate how to maintain classifier effectiveness in distinguishing between targeted and untargeted traffic when all traffic is subject to protection mechanisms.

Robustness. Traffic representations should be resilient to changes in Tor traffic without requiring substantial modifications to their initial design. Such changes may take various forms, including protocol upgrades, content updates, and fluctuations in network conditions. Statistical representations tend to be highly sensitive to these changes, as even minor variations can render the original representations ineffective. This sensitivity is evident in the quantitative measures of information leakage for different statistical features—features that are most informative for website fingerprinting often vary drastically when the traffic is perturbed.

In contrast, stream-based representations are generally more robust because they abstract traffic directly from the original packet streams, making them less affected by such changes. Current stream representation forms, such as sequences, graphs, and images, often draw inspiration from research advances in other fields, including speech signal pro-

cessing, natural language processing, and computer vision. It is anticipated that more powerful and robust representations will continue to emerge, capable of better capturing the inherent characteristics of encrypted network traffic.

Generalizability. A robust classifier should maintain its effectiveness across different networks and environmental conditions. However, many existing studies assume that pre-trained classifiers will be applied in scenarios with conditions similar to those during training. This assumption gives the classifier an unrealistic advantage, as real-world testing conditions often differ from the training environment.

Transferability. To reduce the complexity and resource consumption associated with frequent retraining of classifiers, it is highly desirable for classifiers to be flexibly transferable across different datasets and tasks. Currently, most supervised classifiers are trained with a fixed set of labels, where the model learns an optimal mapping from labeled samples (such as traffic traces or features) to their corresponding labels. During prediction, the classifier outputs a probability vector for each given unlabeled sample. In practice, however, greater transferability is required to design a truly out-of-the-box classifier capable of adapting to novel conditions, unseen datasets, and dynamic real-world environments.

C. The challenges of WF defenses.

In Section V, we sequentially presented the defenses against prevailing attacks as well as those specifically targeting Tor. A clear observation is that defenses in anonymity networks often lag behind emerging attacks. This phenomenon likely arises because defenders are frequently unaware of what threats to anticipate in advance. In Section V-B, we reviewed multiple defenses [7], [35], [37] against WF attacks that have been introduced successively over time. We also observed that anonymity analyses are often conducted under specific, constrained conditions. Comprehensive conclusions, therefore, can typically only be drawn when a wider range of real-world factors is incorporated into the analysis methods. This situation serves as a reminder for network designers and operators to exercise caution and employ rigorous methodologies in formal security assessments. To provide more robust defenses for anonymity networks, it is thus necessary to address the following aspects.

1) *Trade-Offs Under Multiple Factors:* Technically, defense schemes proposed for anonymity networks often involve trade-offs. WF defenses, for example, typically need to balance fingerprinting accuracy, bandwidth overhead, and communication latency. This presents a challenge for designers and operators striving to build scalable, low-latency anonymity networks. Similarly, designing an ideal path selection algorithm that achieves load balancing, defends against both passive and active state-level adversaries, and minimizes information leakage is difficult to realize. For instance, Sun *et al.* [204] proposed a solution to defend against BGP hijacking and interception attacks. Later, Hanley *et al.* [205] improved this scheme by considering worst-case information leakage. However, Mitseva *et al.* [206] raised concerns about the effectiveness of the solution in [205] for defending against BGP attacks and

reducing information leakage. In such scenarios, adaptive and tunable defense schemes may offer a practical alternative.

2) *Unreliable Evaluations of Defense Schemes:* We have also observed the problem of unreliable evaluations of defense schemes within anonymity networks, particularly regarding weaknesses in WF defenses. Notably, many WF defenses overlook adversarial training, which is a critical factor. If a defense scheme is deterministic, an adversary with relevant knowledge could potentially reverse-engineer it. Furthermore, many WF defenses lack comprehensive evaluation metrics and often ignore important factors such as information leakage [207].

3) *Countering Traffic Analysis Attacks in Low-Latency Anonymity Networks:* Theoretically, a WF defense introduces high latency. However, imposing such delays is often impractical in real-world applications. Additionally, as discussed in Section V-B, many existing WF defense schemes suffer from high bandwidth and latency overheads or face challenges in real-world deployment. In addition, we need to balance obfuscation effectiveness with practical deployment costs. We outline two possible countermeasures:

- **Theoretical Perspective:** To build more robust defenses, it is essential to strengthen theoretical foundations—particularly those grounded in deep learning itself. One promising direction is to demystify the black-box nature of deep learning, which can facilitate the development of more fine-grained and targeted defense strategies.
- **Practical Perspective:** Defenses against traffic analysis attacks in anonymity networks can be implemented at multiple levels, namely the network, protocol, and application layers—offering flexibility and the potential for layered protection.

D. Future works

There is still a gap between current research and real-world deployment. As discussed before, the field faces several limitations, including overly strong assumptions, severe concept drift issues, and insufficient data. Addressing these issues is crucial for improving the practicality and reliability of website fingerprinting models in real-world environments.

1) *Relaxing Identification Assumptions:* First, relaxing identification assumptions is a key direction for future website fingerprinting attack research. By reducing assumptions about user behavior, attacker capabilities, and webpage characteristics, while taking more real-world factors into account, the practicality and robustness of website fingerprinting models can be improved to better meet real-world application needs. Second, for user behavior assumptions, future research could focus on designing more flexible user behavior models. These models should allow for realistic browsing behaviors, such as opening multiple browser tabs simultaneously and performing background activities, to more accurately simulate real user behavior. Third, for attacker behavior assumptions, future studies can explore more diverse and complex fingerprinting attack strategies, potentially incorporating simulation-based training in realistic user environments. Fourth, for webpage assumptions, future work could expand to include a wider variety of page types—such as internal pages, frequently updated

pages, noisy pages, and cached pages—thereby enhancing the model’s adaptability and analytical capabilities across different types of web content.

2) *Mitigating the Concept Drift Problem:* In WF models, classifiers are typically trained on pre-collected datasets. However, as time passes, changes in website traffic patterns are not reflected in the trained classifier, leading to a significant drop in classification accuracy. Most existing studies have focused on using deep learning to extract more general and stable fingerprinting features to counteract concept drift, but these methods have reached a performance bottleneck.

Future research can explore model and data update techniques to enable continuous improvement of classifiers. For model updates, approaches such as incremental learning or online learning can be used to incorporate the latest website traffic traces into the training process without retraining the entire model. This improves efficiency and meets real-time update requirements. For data updates, fingerprint data can be continuously monitored and collected. Techniques such as sliding windows can be used to partition the data and regularly discard outdated batches to keep the dataset current. Furthermore, ensemble learning methods can be introduced to build multiple diverse models, reducing the bias and variance of individual neural networks. This can enhance model robustness and improve performance on data affected by concept drift.

3) *Enhancing Data Diversity:* In current datasets used under open-world scenarios, only a limited number of homepages are included, with little to no data collected from internal pages. This limitation can cause website fingerprinting models to fail. In reality, users are free to browse any part of a website, and they are very likely to access multiple different internal pages, even within the same domain. Therefore, future research should involve collecting data from various non-monitored sets and internal pages to evaluate the effectiveness of website fingerprinting models more comprehensively.

Additionally, existing work lacks a systematic study of web caching strategies and often simplifies assumptions by disabling caching altogether. Karunanayake *et al.* [208] found that, under the influence of Tor browser caching, reloading a webpage produces traffic patterns that differ significantly from those generated during the first visit. This finding challenges the effectiveness of current attack models. Thus, future data collection efforts should include webpages with diverse caching strategies, allowing for deeper analysis of how caching behavior affects the performance of WF attacks.

4) *Traffic Segmentation:* Most existing multi-tab WF models require prior knowledge of the number of browser tabs opened on the client side and are trained under the assumption of a fixed number of tabs. When the number of tabs changes dynamically and is not known in advance, these models struggle to perform accurate classification. Future research should aim to relax these constraints and develop more practical and deployable multi-tab fingerprinting models.

Current methods can effectively handle positive and zero time intervals between web page visits. However, for negative time intervals and fully overlapping traffic trace segments, effective segmentation algorithms are still lacking. Future work should explore traffic segmentation techniques that can

handle consecutive or overlapping page visits, which is critical for advancing the real-world applicability of website fingerprinting attacks.

5) *WF Attacks Under Defense Scenarios:* WF attacks still struggle to maintain high accuracy against different types of defenses, including randomization-based defenses, regularization-based defenses, and adversarial defenses. Currently, most defense methods are criticized by users due to added communication overhead and latency, and thus have not been widely adopted by the Tor Project. However, as time progresses, the Tor Project is expected to introduce new defense mechanisms to better protect user privacy. In future research, improving the generality and robustness of attack models to overcome current defense mechanisms remains an important challenge.

6) *Large Language Models (LLMs) in WF attacks and defenses:* LLMs are advanced deep learning models trained on massive text corpora to understand and generate natural language. Built on transformer architectures, LLMs capture complex contextual and sequential dependencies, enabling them to generalize across diverse data modalities beyond text. Recently, researchers have explored applying LLMs to WF [209], [210], where encrypted traffic traces are represented as symbolic sequences. In the LLM-based paradigm, encrypted traffic traces are converted into symbolic or textual sequences that capture packet-level or burst-level characteristics. LLMs can be integrated into WF tasks in multiple ways:

- **LLM-based Attacks:** Pretrained or fine-tuned transformer architectures learn to classify websites directly from tokenized traffic sequences. The models leverage attention mechanisms to capture long-range dependencies between bursts, mimicking how websites load complex resources across time.
- **Prompt-based or Few-shot Learning:** Instead of fully retraining, LLMs can perform WF by conditioning on traffic patterns described through chain-of-thoughts or in-context learning. This approach enhances adaptability to new datasets or defenses without costly retraining.
- **LLM-based Defenses:** On the defensive side, LLMs can simulate potential attacker reasoning or predict how traffic perturbations (e.g. padding, timing noise) affect classification accuracy. Additionally, generative LLMs can design adaptive obfuscation strategies, learning to perturb traffic while preserving utility.

VII. CONCLUSION

With the ongoing advancement of network encryption technologies, ensuring Internet security and privacy remains a critical challenge. Tor, as one of the most widely used anonymity networks, provides essential privacy protection services for users, yet it has also increasingly become a platform exploited for illicit online activities. This survey reviews and analyzes recent research on Tor WF attacks, categorizing existing methods into traditional machine learning-based approaches and deep learning-based approaches. Furthermore, it provides a comprehensive review and analysis of defense strategies against Tor WF attacks, evaluating and comparing these defenses across

multiple dimensions. Finally, this work highlights the current limitations of both Tor WF attacks and defense mechanisms and outlines potential directions for future research aimed at enhancing the security and resilience of anonymity networks.

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